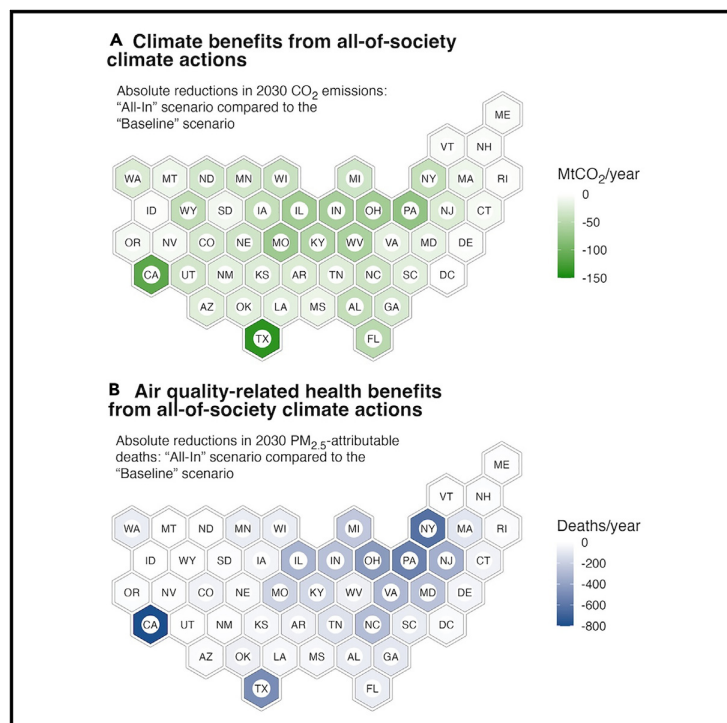


Substantial air quality and health co-benefits from combined federal and subnational climate actions in the United States

Graphical abstract



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In brief

Climate actions not only reduce carbon emissions but also lower the health damage from air pollution. This study finds substantial air quality and health co-benefits from combined federal and subnational climate actions in the United States. Nationally, 6,600 avoided deaths are estimated for 2030, thanks to the reduced exposure to ambient particulate matter. Improved health outcomes are found in all states and nearly all counties. These findings highlight the potential synergy between climate mitigation and human health improvement.

Highlights

- Combining US federal and subnational actions yields widespread health co-benefits
- Climate policies can reduce 6,600 PM_{2.5}-related deaths across the US in 2030
- Health co-benefits are found in nearly all US counties (>99%)
- Besides climate actions, socioeconomic trends and air-quality policies also matter



Article

Substantial air quality and health co-benefits from combined federal and subnational climate actions in the United States

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SCIENCE FOR SOCIETY The burning of fossil fuels, such as coal and gasoline, releases greenhouse gases and air pollutants into the atmosphere, which contribute to climate change and have a detrimental effect on human health. Policies that encourage a transition to cleaner energy sources could help combat climate change, improve air quality, and protect public health at the same time. However, with many different stakeholders involved (such as federal, state, and local governments), each making various policy choices involving subsidies, mandates, and regulations, figuring out how to effectively combine these actions to maximize health benefits remains a challenge. Our findings suggest that a comprehensive approach involving everyone—federal policies alongside actions from states, cities, businesses, and communities—can significantly reduce premature deaths caused by air pollution across the United States. This can improve health outcomes by up to 21% in every state and up to 29% in nearly all counties by 2030. The most effective measures will vary by region, depending on local energy sources and the demographics of the population affected by pollution.

SUMMARY

Policymakers increasingly recognize the air quality and health co-benefits from climate mitigation efforts, yet it remains unclear which combination of policy actions can bring more health co-benefits with more equitable distributions. By developing a coupled energy-pollution-health modeling framework, we show that by 2030, combined efforts from all societal actors across the United States can lower premature deaths attributable to exposure to ambient particulate matter by 6,600 cases (confidence interval: 3,200–10,000) relative to a “baseline” scenario. Net health co-benefits are found in every state and nearly every county (>99%). Future trends in socioeconomic patterns and air-pollution-control efforts are key determinants of the baseline pollution and health levels without climate actions and, consequently, the scale of health co-benefits attributable to climate actions. Our study thus demonstrates the complex and interacting drivers for future health impacts and highlights the importance of coordinated efforts across various policy domains and actors.

INTRODUCTION

The United States (US) has set ambitious targets to address climate change. It aims to cut net greenhouse gas (GHG) emissions by 50%–52% by 2030 from 2005 levels¹ and

achieve net-zero emissions by 2050.² Actions to reduce GHG emissions can also yield health co-benefits by reducing the air pollution from fossil-fuel uses. Studies have found potentially large health co-benefits from climate actions in the US. Nationally, it is estimated that 3,400–20,000 cases of



premature deaths can be avoided in 2030 as the climate and clean energy policies lower the future levels of ambient fine particulate matter (PM_{2.5}).^{3–7} The variations in the magnitude of the co-benefits depend on the design and stringency of climate policy,^{5,8} the complexity of the atmospheric processes being modeled,^{9,10} relevant sociodemographic projections,¹¹ and the choice of exposure-response relationships used for health assessment.^{12–14}

While the potential for health co-benefits is generally acknowledged, there are at least two important policy-relevant questions that remain understudied. First, climate-policy efforts can come from a wide range of societal actors, including federal, state, and local governments, as well as businesses and community organizations.¹⁵ Prior studies have evaluated the health co-benefits of isolated actions, such as federal policies (e.g., carbon tax, cap-and-trade, and emission standards),^{4,6,16–20} state-led initiatives (e.g., renewable portfolio standards and energy efficiency mandates),^{5,21–25} and local actions taken by cities and communities.^{26,27} Some studies have also examined optimal energy-system and climate-policy design to maximize air quality and health co-benefits.^{28–32} A systematic assessment that captures the interactions of federal and subnational actions across various sectors is needed to improve the decision relevance of co-benefit analysis.

Second, the future air quality and health landscape is also affected by co-evolving factors outside the scope of climate actions (Figure 1A). For instance, tightening the air-pollution-control efforts on fossil-based facilities and technologies can mitigate air pollution without shifting to lower-carbon energy sources.^{33,34} Furthermore, socioeconomic trends influence the health risks and damages from future air-pollution exposure, as they affect not only the overall energy demand and the challenge for decarbonization but also the size and vulnerability of the exposed populations.^{11,35} These factors largely determine the baseline pollution and health trends in the absence of climate actions, which play an important role in determining the magnitude of health co-benefits that can be achieved by additional climate actions. There remains a knowledge gap in understanding how these non-climate drivers interact with climate actions to shape the scale and distribution of health co-benefits.

How will the combined climate-policy actions from federal and subnational actors influence future air quality and health in the US? Motivated by this policy-relevant question, we aim to address the aforementioned knowledge gaps by examining the co-benefits of combining the climate actions from the actors at all levels across the whole country. From the method perspective, we develop a multi-actor, multi-scale modeling framework to couple the modeling of energy, emissions, air pollution, and health impacts. This framework enables the assessment of individual and combined climate actions and understanding of their interactions with other non-climate drivers. From the policy insight perspective, we demonstrate that health co-benefits from climate actions can be large and widespread, but the magnitude of the co-benefits is affected by a range of factors, including the choice and implementation of climate policies as well as the future trends in sociodemographic factors and air-pollution-control efforts.

RESULTS

Methods summary

To capture key mechanisms shown in Figure 1A, we develop a coupled energy-pollution-health modeling framework³⁶ (Figure 1B) by linking four components: (1) a state-level integrated assessment model to assess energy and emissions (i.e., the Global Change Analysis Model with state-level representation for the US,³⁷ GCAM-USA); (2) a downscaling module to estimate grid-level emissions (12 km × 12 km) based on the 2017 National Emission Inventory (NEI)³⁸; (3) a reduced-complexity air-quality model, Intervention Model for Air Pollution (InMAP),³⁹ to simulate annual ambient concentrations of PM_{2.5}; and (4) a county-level health-impact assessment based on the Environmental Benefits Mapping and Analysis Program (BenMAP)¹² to quantify the premature deaths attributable to long-term PM_{2.5} exposure.

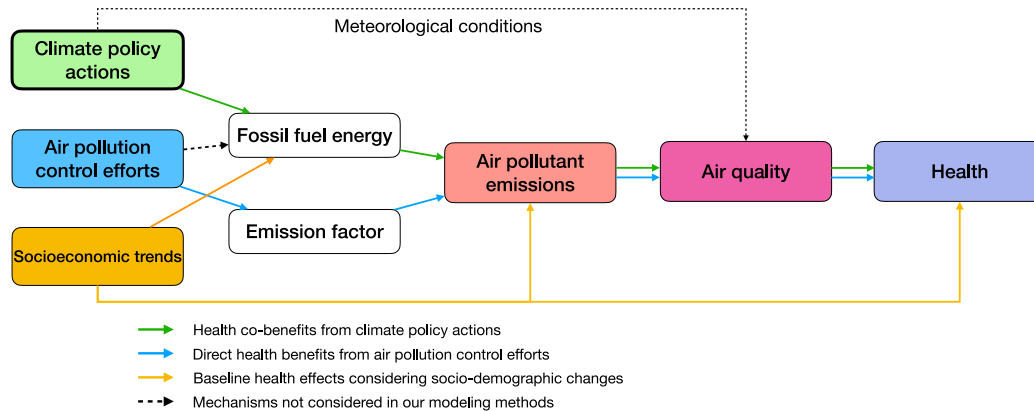
This framework allows us to represent some of the key mechanisms laid out in Figure 1A. For instance, climate-policy actions and socioeconomic trends collectively shape the energy use patterns, while the pollution-control efforts affect the emission factors of relevant energy activities. The resulting emissions of air pollutants influence local air quality. The health outcomes are further determined by size and vulnerability of the exposed population and their exposure level. We recognize this framework still neglects some other mechanisms (e.g., effects of a changing climate on pollution formation and transport), exposure pathways (e.g., ozone exposure), and health endpoints (e.g., short-term morbidity impacts), which are addressed in the discussion.

Using this modeling framework, we simulate and compare three scenarios from 2015 to 2030 (for more details, see Tables 1 and 2)³⁶.

- (1) A “baseline” scenario that considers the effects of socioeconomic changes and business-as-usual pollution-control efforts in the absence of climate-policy efforts
- (2) An “existing climate policies” scenario that considers current climate and clean energy policies including the financial incentives in the federal Inflation Reduction Act (IRA) and relevant state initiatives such as the renewable portfolio standards and electric vehicle (EV) sales targets
- (3) An “all-in climate policies” scenario that represents a high-ambition, all-of-society approach including enhanced climate actions from the federal government and states, and also supportive actions from cities, businesses, and other non-federal actors.^{1,40} Here we identified a portfolio of actions by different actors that can keep the US on track to reaching its 2030 climate target (i.e., a ~52% reduction in GHG emissions by 2030 relative to 2005; for more details see Zhao et al.⁴⁰)

The differences between the two climate-policy scenarios compared to the “baseline” scenario help quantify the health co-benefits from climate actions. We further develop a sensitivity analysis to demonstrate how changes in the “baseline” scenario (i.e., socioeconomic trends and pollution-control efforts) might influence the magnitude and distribution of health co-benefits attributable to climate actions.

A Conceptual framework



B Integrated modeling methods used in this study

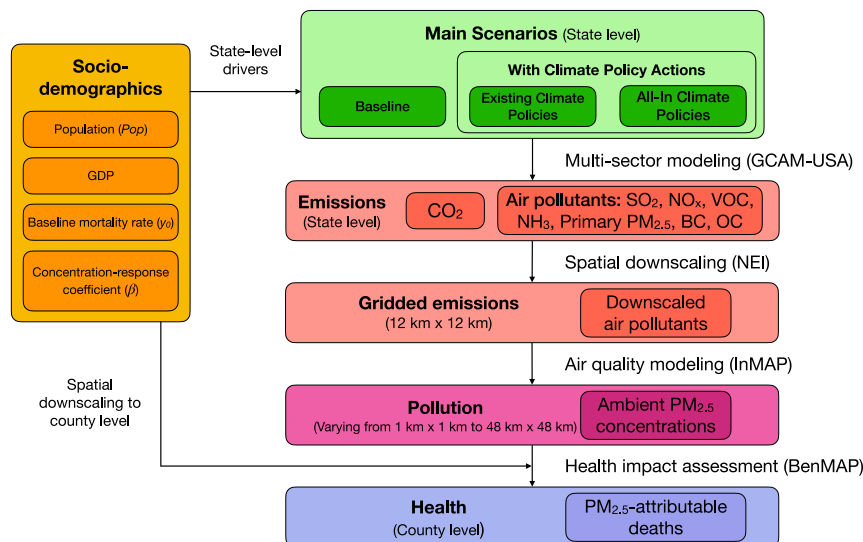


Figure 1. Conceptual framework and modeling methods used in this study

(A) Here, we distinguish health co-benefits from climate-policy actions, direct health benefits from air-pollution-control efforts, and the baseline health effects considering sociodemographic changes.

(B) Our “baseline” scenario considers the effects of sociodemographic trends and the business-as-usual air-pollution-control efforts. We then consider two climate-policy scenarios that represent current and enhanced policy efforts from federal, state, and local actors. Variations in the “baseline” scenario and the resulting co-benefit assessment are shown in Figure 6.

See the Methods section and Table S1 for definitions of related terms.

Nationwide co-benefits from all-in climate actions

For climate outcomes, we find that climate policies significantly reduce CO_2 emissions, with deeper reductions under more ambitious scenarios. In the “baseline” scenario, we estimate the 2030 national CO_2 and GHG emissions, respectively, to be 21% and 11% lower than 2015 levels or with 31% and 20% reduction relative to 2005 levels, which is the baseline year for the US climate target (4.1 and 5.4 GtCO_2eq GHG in 2030, compared to 5.2 and 6.1 GtCO_2 in 2015 and 6.0 and 6.7 GtCO_2eq in 2005). This suggests that considering the moderate growth projected for population and energy demand, the anti-

ipated changes in the energy system, such as efficiency improvements and a shift away from coal, are on track to lower US GHG emissions independent of additional climate actions. In comparison, relative to 2015, we estimate 34% and 46% reductions in national CO_2 emissions in the “existing climate policies” and “all-in climate policies” scenarios, respectively, associated with 31% and 41% reductions in nationwide GHG emissions. This confirms that substantial GHG mitigation potential from climate actions exist in all major sectors, especially electricity and transportation sectors (for more details see Zhao et al.⁴⁰; see also Figure S1 for changes in national emissions by sector).



Table 1. Summary of the “baseline” and the “existing climate policies” scenarios and relevant policies

Scenario	Existing climate policies		
	Baseline	Federal	State
Sectoral climate policies	power	production tax credit and investment tax credit for renewables; energy infrastructure reinvestment program; scheduled coal plant retirements	binding renewable portfolio standards
	industry	45Q tax credit and hydrogen tax credit to increase use of clean fuels and CCS; manufacturing investment tax credit for advanced energy projects, which increase electrification for low-temperature heating projects; advanced industrial facilities deployment program, which is assumed to result in faster stock turnover for fossil fuels	N/A
	building	rebates and incentives to increase efficiency in residential and commercial buildings	Energy Efficiency Resource Standards (EERS)
	transportation	clean vehicle tax credits; Corporate Average Fuel Economy (CAFE) standards	EV sales targets for passenger and freight vehicles, following California's Advanced Clean Cars II (ACC II) and Advanced Clean Trucks (ACT)
	land	investments in forest management, land conservation, and ecosystem restoration	N/A

For more details, see Zhao et al.¹



Table 2. Summary of the “all-in climate policies” scenarios and relevant policies

Scenario	All-in climate policies ^a			States	Cities and businesses
	Federal				
Sectoral climate policies	power	proposed EPA regulations on coal and gas plants; coal generation phaseout by 2030	accelerated clean energy standards targeting 80% of electricity demand by 2030; coal generation phaseout by 2030	city-wide clean electricity goals; coal generation phaseout by 2030	
	industry	heightened sequestration levels from 45Q tax credits; no new coal in industry	all states with cement production install CCS capability for 40% of cement produced by 2035, consistent with California’s 40% goal	“buy clean” targets; accelerate deployment of green technologies	
	building	N/A	heightened adoption of stringent EERS standards; zero-emissions appliance standards for water and space heating	accelerating energy savings and phaseout of gas appliances; building performance standards; expanded voluntary target setting	
	transportation	improved CAFÉ standards; cash for clunkers program to retire old, inefficient vehicles; 100% federal fleet procurement of zero-emission light duty vehicles (LDVs), light commercial trucks, and buses by 2030	widespread adoption of ACC II and ACT; vehicle miles traveled (VMT) reduction policies	VMT reductions through planning, low- and zero-emissions zones, and decongestion pricing; major auto manufacturers on track to 100% ZEVs for new LDV sales by 2035	
	land	expanded investment in wildfire risk mitigation	targeted programs and investment in reforestation, soil carbon sequestration, climate-friendly agricultural practices, and wildfire mitigation; increased investment in GHG quantification and monitoring	expanded urban forestry efforts and fire-management practices; increased investments in land-based climate mitigation strategies	

For more details, see Zhao et al.¹

^aAll the policies that are listed under “existing climate policies” are also implemented in the “all-in climate policies” scenario.

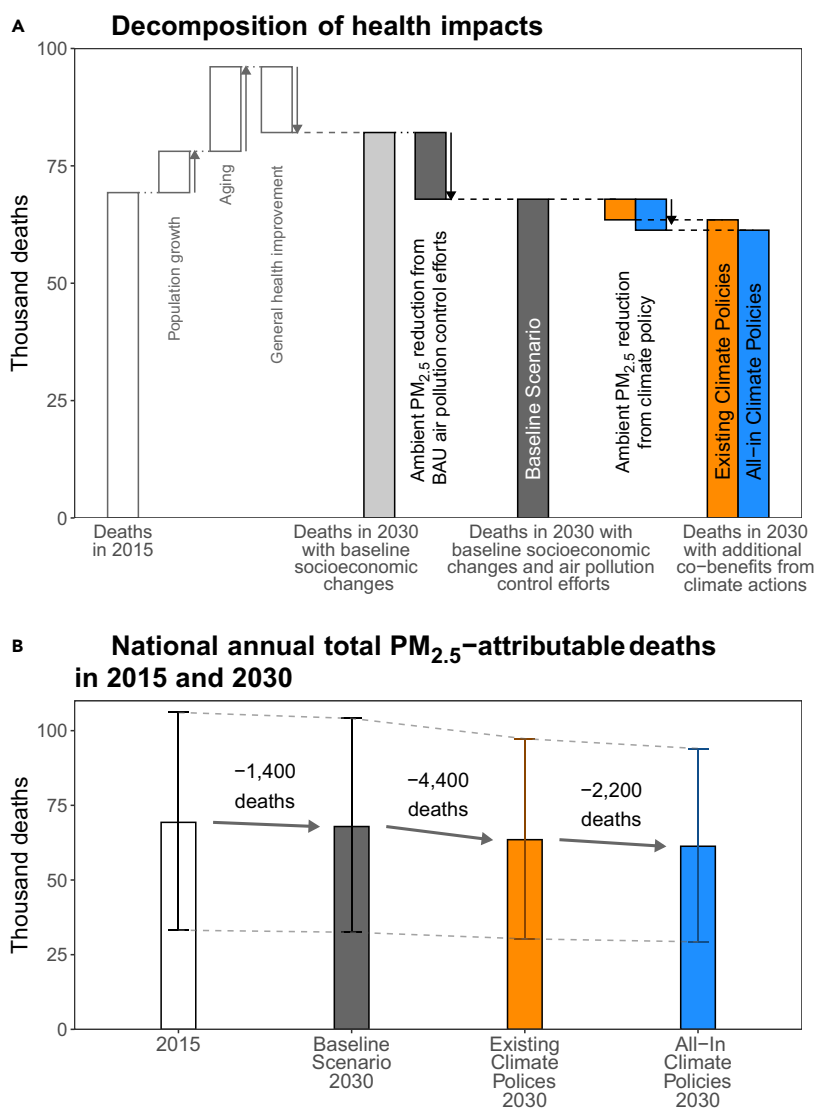


Figure 2. Changes in future nationwide PM_{2.5}-attributable deaths and underlying drivers

(A) The impacts of four individual factors on the aggregate changes in PM_{2.5}-attributable deaths: population growth, population aging, general health improvement (measured by baseline mortality rates), and the exposure level to ambient PM_{2.5}. The first three factors are the same across three policy scenarios, whereas the last one varies between “baseline” and the climate-policy scenarios. The light-gray bar indicates the results considering only the health effects of socioeconomic factors. The dark-gray bar indicates the results of the “baseline” scenario with combined effects of the socioeconomic factors and business-as-usual (BAU) air-pollution-control efforts. The orange and blue bars show the results for the two climate-policy scenarios.

(B) PM_{2.5}-attributable deaths in 2015 and three 2030 scenarios: “baseline,” “existing climate policies,” and “all-in climate policies.” The bars and arrows show the deaths in each scenario and the changes between them. Error bars indicate the 95% confidence interval (CI) based on the CI of the exposure-response functions.⁴⁵

the baseline trends in both socioeconomic factors and pollution-control efforts, we estimate the 2030 PM_{2.5}-attributable deaths in the main “baseline” scenario to be 67,900 (CI: 32,500–104,100), which is 2% lower than in 2015 (Figures 2A and 2B).

Importantly, additional climate-policy actions can yield substantial air quality and health co-benefits (Figure 2). Nationally, comparing the “existing climate policies” scenario to the “baseline” scenario, we project ~4,400 (CI: 2,200–6,800) fewer PM_{2.5}-attributable deaths in 2030 (Figure 1A). Under the “all-in climate policies” scenario, we find a

For the health outcomes, we first confirm that sociodemographic trends and pollution-control efforts can affect the baseline pollution and health levels in the absence of climate-policy efforts. In 2015, we estimate a national total of ~69,300 ambient PM_{2.5}-attributable deaths (confidence interval [CI] based on exposure-response coefficient: 33,200–106,100; Figure 2B). This magnitude is in line with prior studies (range: 30,000–200,000).^{41–44} In 2030, considering only the effects of sociodemographic factors, we estimate the PM_{2.5}-attributable deaths to increase by 12,800 compared to 2015. This is a combined effect of (1) population growth and aging that increase the size and vulnerability of future populations and (2) improvements in the general health conditions (measured as decline in baseline mortality rate) that lower the vulnerability of future populations. When further adding the effects of business-as-usual air-pollution-control efforts, we find the lowering of the ambient PM_{2.5} level leads to a reduction in 14,200 PM_{2.5}-attributable deaths. Combining

further reduction of ~2,200 (CI: 1,000–3,300) deaths in 2030, resulting in a total of ~6,600 (CI: 3,200–10,100) fewer deaths in 2030 relative to the “baseline” scenario (Figure 1B). This comparison highlights that current climate policies are able to bring substantial health co-benefits, while enhancing the policy ambition from actors across all of society can further increase co-benefits.

Since the choice of exposure-response relationship is found to be important for health assessment,^{12,13} we complement our main results that are based on Krewski et al.⁴⁵ with a sensitivity analysis using the estimate from the Global Exposure Mortality Model¹³ (GEMM). We find that using GEMM results in a higher estimate for PM_{2.5}-attributable deaths in all cases, i.e., 205,000 (CI: 123,000–286,000) in 2015, 190,000 (CI: 115,000–265,000) in 2030 “baseline” scenario, 173,000 (CI: 105,000–241,000) in 2030 “existing climate policies” scenario, and 165,000 (CI: 100,000–230,000) in 2030 “all-in climate policies” scenario.

Our main insights remain robust, i.e., that the baseline trends influence the future health outcomes and the health co-benefits from climate policy are likely to be substantial.

Varying spatial distributions of the health co-benefits

We find that the emissions and health effects are distributed unevenly across states and counties. When co-benefits are measured by the absolute changes in PM_{2.5}-attributable deaths in the climate-policy scenarios compared to the “baseline” scenario, greater co-benefits are found in states with larger economies and populations. For instance, larger emitters such as California, Texas, New York, and Pennsylvania are projected to reduce more CO₂ emissions and more PM_{2.5}-attributable deaths (Figures 3A and 3B; see state-level maps in Figure S2). This confirms the importance of population and economy size in determining the overall size of the emissions and health outcomes.

In comparison, when co-benefits are measured by the relative changes in PM_{2.5}-attributable deaths, we observe greater cross-state variations, which reflect the unique state features in energy structure, climate-policy efforts, and sociodemographic patterns such as aging. Quantitatively, comparing the “existing climate policies” scenario to the “baseline” scenario, state-level CO₂ emissions in 2030 decrease by 8%–56% across 50 states and Washington DC. These climate actions bring concurrent air quality and health co-benefits in all states. Across the country, the state-level concentrations of ambient PM_{2.5} decrease by up to 13% (Figure 3C) and the PM_{2.5}-attributable deaths decrease by 0.5%–13% (Figure 3E). The health co-benefits, measured as percent changes, are the greatest in the Northeast, Midwest, and South regions (based on the Census Regions and Divisions of the United States). With additional climate actions under the “all-in climate policies” scenario, we find greater reductions in CO₂ and health co-benefits in every state. Comparing the “all-in climate policies” scenario to the “baseline” scenario, state-level CO₂ emissions in 2030 decrease by 10%–100% across the country. These climate benefits are accompanied by additional clean air and health co-benefits: the state-level concentrations of ambient PM_{2.5} decrease by 1%–22%, and the PM_{2.5}-attributable deaths decrease by 2%–21% (Figure 3D). The largest percentage reductions are found in West Virginia (21% reduction), Kentucky (19% reduction), and Maryland (18% reduction) (Figure 3F; see Figure S3 for changes in energy consumption in these states).

Notably, the relationship between the relative changes in CO₂ and PM_{2.5}-attributable deaths varies across states. For instance, comparing the “existing climate policies” scenario to the “baseline” scenario, New York and Wisconsin are expected to reduce their CO₂ emissions by 20% and 27%, respectively. However, PM_{2.5} concentrations in New York are reduced by 9%, but Wisconsin has only a 5% reduction. This state-level variation reflects the differences in energy structure in response to climate policies: New York is projected to shift from traditional biomass and gas to clean energy in the building sector, which significantly reduces the emissions of particulate matter from biomass combustion.⁴⁶ In comparison, Wisconsin, already more electrified in the building and transport sectors even without climate actions, is projected to reduce its coal power, which is less polluting than biomass burning and therefore realizes lower improvement in PM_{2.5} concentration (see Figure S4 for the changes in energy consumption in New York, Wisconsin, and other states for com-

parison; see Table S3 for energy consumption and CO₂ emissions in all states).

Diving deeper into the county-level impacts, we find substantial health co-benefits in many counties across the country, measured both by relative changes (Figure 4) and absolute changes (Figure S5). For instance, comparing the “existing climate policies” scenario to the “baseline” scenario, 43% of counties are expected to decrease their PM_{2.5}-attributable deaths by more than 5% (Figure 4B). However, 2% of counties, primarily in Nevada and Idaho, are projected to have small increases of PM_{2.5}-attributable deaths in the “existing climate policies” scenario (<1%). This is because these counties have low baseline pollution levels and therefore limited mitigation potential from climate actions. These counties are therefore sensitive to increases in background pollution levels, including from increased nationwide agricultural activities as a result of growing demand for bioenergy crops such as corn and soybean.

With additional climate actions under the “all-in climate policies” scenario, nearly all the counties are expected to experience net health co-benefits compared to the “baseline” scenario. Among 3,108 counties in the continental U.S., 67% of the counties expect reductions of more than 5%, and 31% of the counties expect reductions of more than 10% (Figure 4C). The additional benefits, measured by absolute changes, are particularly large in the Greater Los Angeles area, the San Francisco Bay area, and the Seattle metropolitan area. This is a combined effect of high population density in these areas and a large reduction in ambient PM_{2.5} concentrations thanks to the federal, state, and other subnational policies that can collectively bring down the pollution levels in these regions.

Determinants of state-level health co-benefits

Climate actions influence human health through a complex cause-effect pathway, from energy and emissions to air quality and health. To demonstrate the complexity and interactions of these drivers, here we dive into six selected states as examples and compare the changes in the “all-in climate policies” scenario with those in the “baseline” scenario. These six states vary in important features that determine the potential for co-benefits, including California and Texas, as large economies and emitters, and Ohio, North Carolina, Virginia, and Arizona, as a range of states with varying sizes of economy, energy structures, current emissions, and climate-policy ambitions (Figure 5).

First, the transition away from coal and gasoline is the leading driver of emissions reductions, although some low-carbon energy technologies may add new sources of air pollutants. For the power sector, the transition from coal to renewable energy is the driving force behind the reduction of SO₂ and nitrogen oxides (NO_x) emissions (Figures S7 and 5A). For the transportation sector, most states are expected to reduce NO_x emissions, thanks to the shift away from gasoline and diesel-based vehicles to those powered by electricity or secondary biofuels. For the industrial sector, states that transition from coal to electricity are expected to reduce more SO₂ emissions than those that transition from coal to refined liquids or natural gas. For the building sector, less reduction in absolute emissions is found compared to the other sectors due to its lower contribution to national emissions. However, substantial percent reductions are found in this sector, including a 1%–29% reduction in SO₂, a 3%–31%

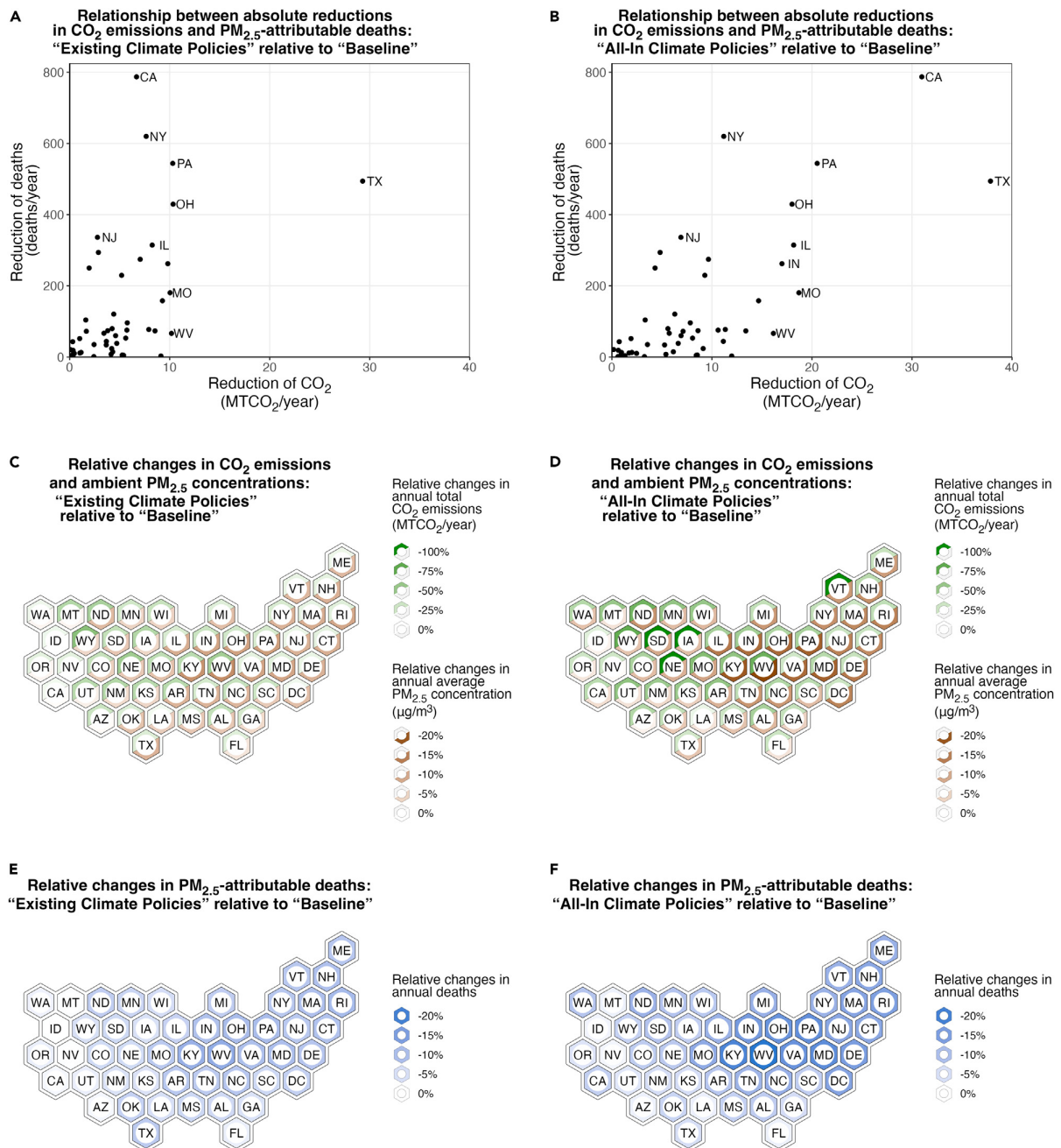


Figure 3. State-level reductions in CO₂ emissions, PM_{2.5} exposure, and PM_{2.5}-attributable deaths in 2030

(A and B) Absolute changes in the two climate-policy scenarios compared to the “baseline” scenario; here each state is represented as a dot in the scatterplot. States with large reductions in CO₂ or large reductions in deaths are labeled with state names.

(C–F) Relative changes in the two climate-policy scenarios compared to the “baseline” scenario; here we show state-level maps to visualize the spatial patterns more directly. See Table S2 for the values of relative and absolute changes in PM_{2.5}-attributable deaths in the two climate-policy scenarios by state.

reduction in NO_x, and a 7%–20% reduction in primary PM_{2.5} emissions across the six selected states. Finally, some sectors and states might experience an increase in emissions in the “all-in climate policies” scenario compared to the “baseline”

scenario. For instance, we observe a large increase in transport-related NO_x emissions in Texas, mainly due to the projected shift from road freight to shipping, combined with limited pollution control assumed for the shipping sector. We also observe

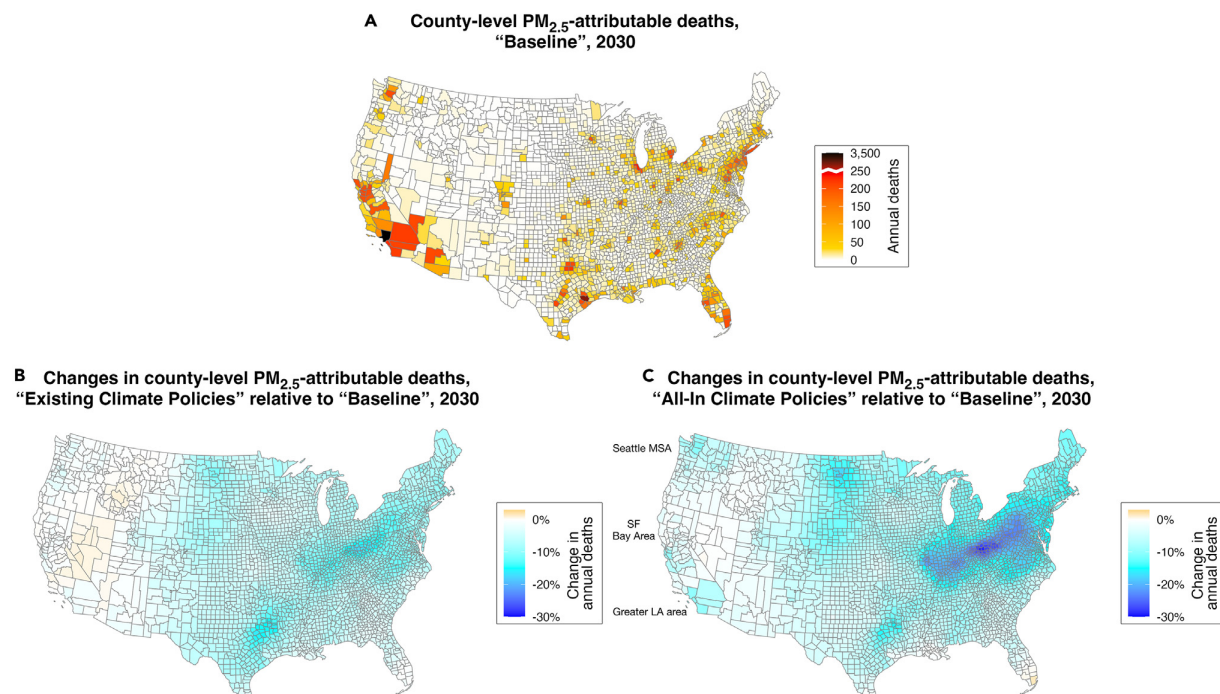


Figure 4. County-level $\text{PM}_{2.5}$ -attributable deaths in 2030

(A–C) (A) County-level $\text{PM}_{2.5}$ -attributable deaths in the “baseline” scenario. (B and C) Percent changes of county-level $\text{PM}_{2.5}$ -attributable deaths under the two climate-policy scenarios.

See [Figure S5](#) for the absolute changes in $\text{PM}_{2.5}$ -attributable deaths. See [Figure S6](#) for county-level attainment of the National Ambient Air Quality Standards (NAAQS) for $\text{PM}_{2.5}$ ($9 \mu\text{g}/\text{m}^3$).

a slight increase in electricity-related SO_2 emissions and primary $\text{PM}_{2.5}$ emissions in California. This is primarily due to an increase in the generation from bioenergy plants with CCS, together with a slight increase in generation from coal with CCS plants. These potential increases in air-pollutant emissions due to additional climate actions are also found in other studies,^{46,47} which highlights the importance of concurrent efforts to strengthen pollution-control efforts.

Second, these changes in precursor air-pollutant emissions, especially the huge decline in SO_2 (a per-capita reduction of 0.4–6.8 kg/year) and NO_x (a per-capita reduction of 1.0–3.7 kg/year, excluding Texas) emissions, contribute to a net decline in ambient $\text{PM}_{2.5}$ concentrations in all six states. However, the differences across the states are driven not only by precursor emissions but also by local meteorology and atmospheric conditions. For instance, despite fewer reductions in pollutant emissions than Ohio, Virginia experiences a similar reduction in ambient $\text{PM}_{2.5}$ concentrations. This effect is largely due to cross-state transport of air pollutants, as we demonstrate in a sensitivity analysis where we keep the emissions in Virginia the same as the “baseline” scenario and only reduce the emissions in the five surrounding states (West Virginia, Kentucky, Maryland, Pennsylvania, and North Carolina) to the level in the “all-in climate policies” scenario. We find that emissions reductions in the surrounding states—and the subsequent reduction in pollution transport to Virginia—contribute to 44% of the reductions in $\text{PM}_{2.5}$ concentrations in Virginia and

42% of the reductions in the state-level $\text{PM}_{2.5}$ -attributable deaths ([Figures S8 and S9](#)). This highlights that the health co-benefits can go beyond state borders; pollution reduction in upwind regions can also benefit the downwind regions.

Finally, local sociodemographic characteristics affect population vulnerability, influencing health outcomes. For instance, despite a lower $\text{PM}_{2.5}$ concentration reduction in California compared to Texas, we find a larger health co-benefit in California measured by the reduction in $\text{PM}_{2.5}$ -attributable death rate. This discrepancy is largely a result of demographic trends as we project a higher percentage of the elderly population (aged 65 and older) in California, estimated at 19%, in comparison to 16% in Texas, in 2030. Given that the elderly demographic is particularly vulnerable to the adverse effects of air pollution, this factor influences differential health co-benefits observed between the two states.

In summary, these state-varying results underscore the complex interactions of climate actions, energy transitions, cross-state pollutant transport, and sociodemographic patterns. Furthermore, they highlight the importance of considering state-specific characteristics when designing health-oriented climate action to protect local communities.

Effects of policy enforcement and coordination

Here, we demonstrate the importance of key policy considerations on the assessment of co-benefits. First, we find that

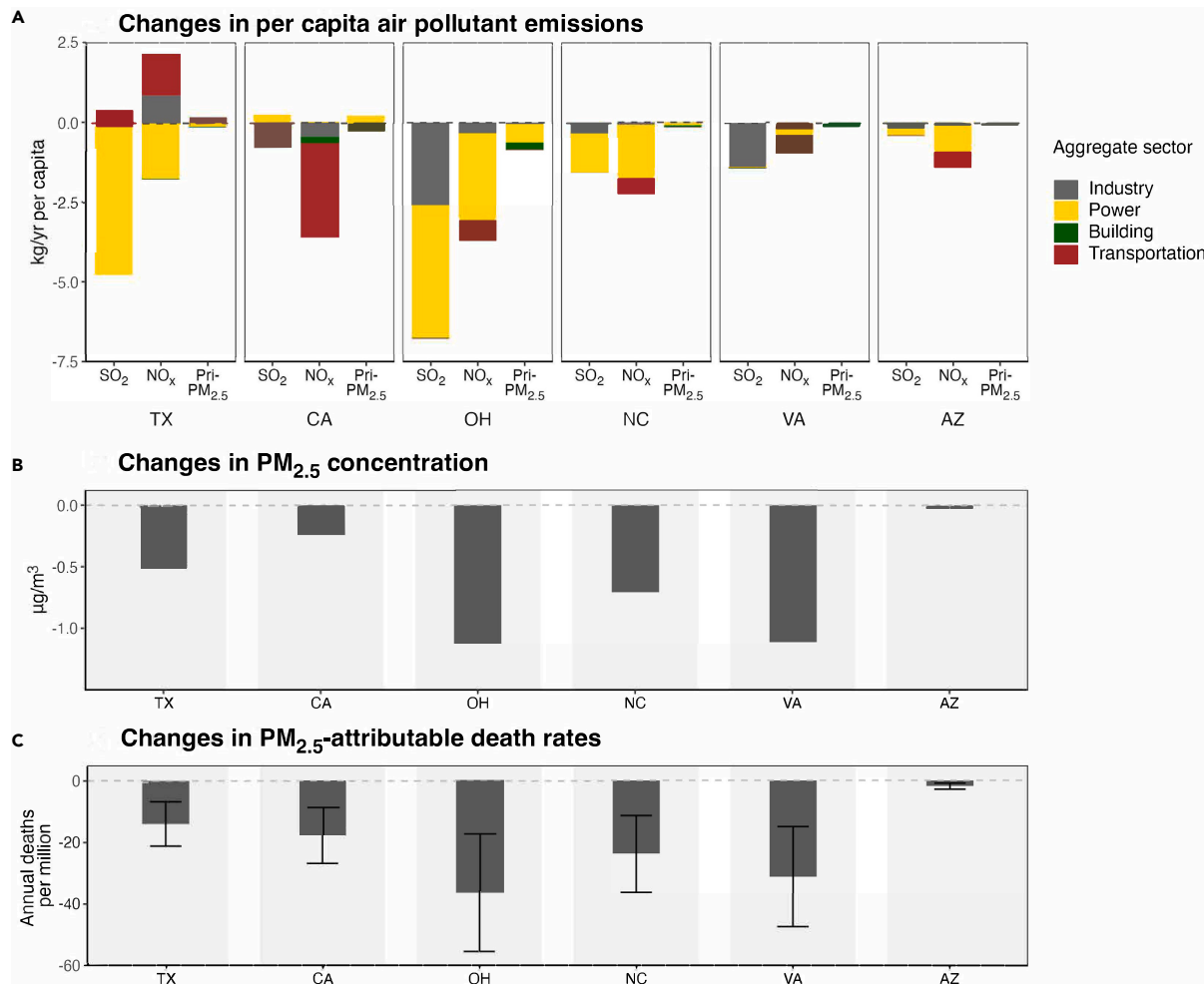


Figure 5. State-level changes in sectoral air-pollutant emissions, ambient PM_{2.5} concentrations, and health impacts in 2030 in the “all-in climate policies” scenario relative to the “baseline” scenario

(A) The changes in per capita SO₂, NO_x, and primary PM_{2.5} emissions by aggregate sectors at the state level.

(B) The changes in ambient PM_{2.5} concentrations.

(C) The changes in PM_{2.5}-attributable death rate as a measure for health risks.

Error bars indicate the 95% CI based on the CI of the exposure-response functions.⁴⁵

incomplete enforcement of federal climate and energy policies can substantially reduce the health co-benefits. For instance, if we assume only 80% of the emission reduction potential is realized due to imperfect enforcement, we find the co-benefits, measured by percent reductions in PM_{2.5}-attributable death rate in the “all-in climate policies” scenario relative to “baseline,” are reduced by 14%–50% across the six states (Figure 6). This emphasizes the importance of successfully enforcing federal energy and climate policies to realize the full potential of health co-benefits.

Second, the most effective state-level climate action in delivering health co-benefits varies across states. For instance, we perform a sensitivity analysis for two major state-level policies (EV credit and sales targets and coal-phaseout policy) in three selected states (California, Texas, and Pennsylvania) (Fig-

ure S10). We demonstrate that the state-level EV policy has large potential for generating health co-benefits in California, whereas coal-phaseout policy is effective in bringing health co-benefits for Texas. We also find large co-benefits for Pennsylvania from both policies. These results suggest that targeting high-polluting sectors generally yield greater health co-benefits. The choice of state and local climate actions needs to be location specific with considerations of current energy structure and technology stock.

Third, pollution-control efforts and socioeconomic changes influence the baseline health outcomes and thus the magnitude of the health co-benefits from climate actions. For instance, if end-of-pipe controls are implemented to reduce the emission factors of major air pollutants from fossil activities by 20%, we estimate the PM_{2.5}-attributable death rate in the “all-in climate policies”

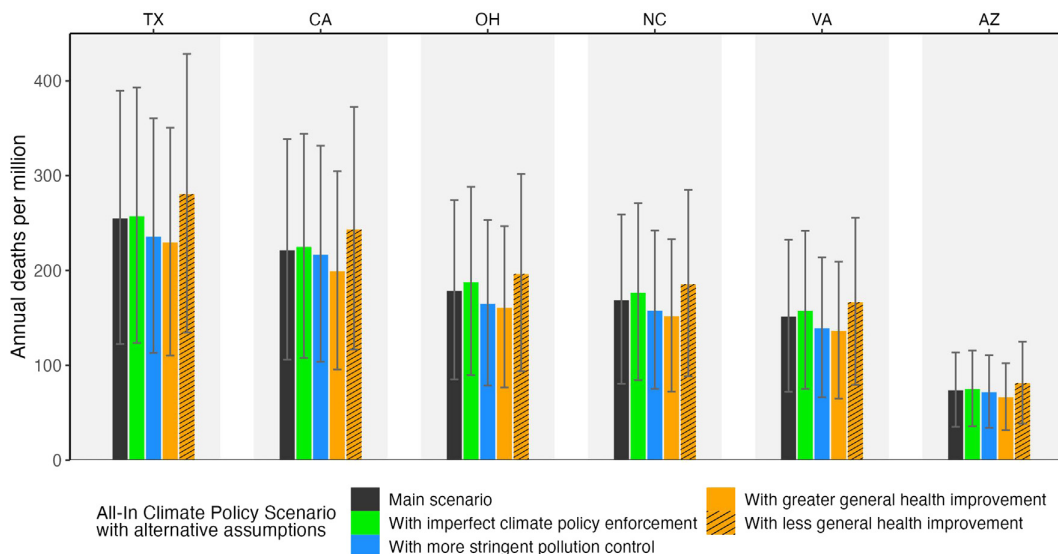


Figure 6. Effects of alternative policy assumptions on the $PM_{2.5}$ -attributable death rate in six selected states in the “all-in climate policies” scenario

Error bars show the 95% CI based on the upper and lower bounds of the exposure-response coefficients. With more stringent pollution control, we assume the emission factors of fossil fuels in the power, industry, building, and transportation sectors are 20% lower than in the main scenario. With the imperfect climate-policy enforcement, we assume only 80% of the emission reduction potential (i.e., main “all-in climate policies” scenario compared to “baseline”) is realized. With greater or less general health improvement, we assume the annual $PM_{2.5}$ -attributable baseline mortality rates are approximately 10% lower and 10% higher than the main scenario in 2030, which follows the SSP1 and SSP3 assumptions compared to SSP2.³⁵

scenario to be reduced by 3%–8% across the six states. In a similar vein, if we assume the decline in baseline mortality rate from 2015 to 2030 is approximately 80% faster (i.e., changing from SSP2 assumption in the main results to SSP1), the $PM_{2.5}$ -attributable death rate in the “all-in climate policies” scenario is reduced by 7%–12% across the six states (Figure 6). However, these improvements in health outcomes are a result of pollution-control efforts and general health conditions independent of climate actions. As fossil technologies get cleaner and the pollution reductions are experienced by a less vulnerable population, the magnitude of the co-benefits from climate policies is reduced.

Finally, the distribution of the health co-benefits and the broader equity implications depend on how equity is being conceptualized and measured. As an illustrative example, we compare counties with varying Social Vulnerability Index (SVI) at present⁴⁸ and consider various metrics to measure co-benefits. When co-benefits are measured by the reductions in $PM_{2.5}$ -attributable deaths, we find counties with higher SVI, on average, experience greater co-benefits. However, when co-benefits are measured by the reductions in $PM_{2.5}$ -attributable death rates to normalize the effects of population size, we find counties with lower SVI are expected to experience greater co-benefits (Figure S11). This demonstrates the importance of carefully operationalizing and measuring equity in policy analysis to understand the equity implications that are most relevant to decision makers and affected stakeholders.^{49,50}

In summary, our sensitivity analyses demonstrate the importance of policy enforcement, selection, and coordination for co-benefits assessment. It implies that strategically coordinating

the efforts across actor types (e.g., federal, state, and local) and policy domains (e.g., energy, climate, social, air quality, and equity) are crucially needed to improve the pollution and health outcomes in the future.

DISCUSSION

Our study finds substantial air quality and health co-benefits in the “all-in climate policies” scenario, where federal and state actors take enhanced actions and city and business actors support their efforts to mitigate climate change and transition to clean energy (see Table 1 for actors and actions). The magnitude and distribution of the health co-benefits are influenced by the choice and enforcement level of climate policy actions. The broader air quality and health outcomes are further affected by socioeconomic trends and air-pollution-control efforts. These changes in non-climate aspects will also play an important role in determining the scale of co-benefits.

Policy implications

We highlight two main policy implications. First, increasing the health co-benefits from climate actions requires strategically pursuing synergistic policy actions that can co-reduce CO_2 emissions and air pollution. For instance, in the power sector, actions that shift generation from coal to wind and solar are particularly effective in reducing SO_2 and NO_x emissions in addition to CO_2 emissions. Synergistic policies include federal tax incentives for renewable electricity, state mandates such as renewable portfolio standards, and local initiatives such as clean electricity targets and community solar projects. In the transportation

sector, federal fuel efficiency standards, state EV incentives, and local EV charging infrastructure investments can promote low-carbon and cleaner mobility. This shift from internal combustion engine to EV reduces both CO₂ emissions and tailpipe air-pollutant emissions such as NO_x and VOC. In the industrial sector, effective strategies include federal grants for clean industrial technologies, state industrial decarbonization policies, and local green manufacturing zones. All of these efforts can facilitate the adoption of renewable energy, increase energy efficiency, and reduce emissions of SO₂, NO_x, and primary PM_{2.5}. In the building sector, federal energy efficiency programs, state building codes for energy performance, and local retrofit initiatives enhance energy conservation, lower the carbon and environmental impacts. Reduced energy consumption in the building sector can also lower the electricity demand, further reducing the emissions of air pollutants from the electricity sector.

Second, coordinating efforts across climate, air quality, and other socioeconomic aspects will be essential to improve the overall pollution and health outcomes. In addition to the co-benefits from climate actions, the pollution-related health damage can be reduced by improving general health conditions and tightening pollution-control efforts. Coordinating policy actions across energy, climate, and health issues offers great potential for reducing future pollution and health impacts. However, it also presents significant challenges due to the decentralized governance structure and diverse policy actors. For instance, energy decision-making often involves the Department of Energy setting overarching energy policies at the federal level, which are then interpreted and implemented by state and local agencies. In the meantime, the air-quality decision-making is spearheaded by the Environmental Protection Agency at the federal level, including setting the National Ambient Air Quality Standards (NAAQS) and other technology- and sector-specific emission standards. State agencies are then in charge of the State Implementation Plan to ensure all counties within the state meet the NAAQS. Since air quality and health are primarily local concerns, enhancing the capacity of states and local actors will be crucial to facilitate coordinated decisions for energy, climate, and health.

Future directions of research

Our analysis also opens up important research questions for future investigation. First, a broader coverage of health-related metrics, ranging from pollution exposure to morbidity and mortality metrics, are needed for a more comprehensive health-impact assessment. Our assessment is limited to the long-term mortality impacts associated with the exposure to ambient PM_{2.5}. Yet there is compelling evidence that other ambient pollutants such as ozone and indoor air pollution may also have important negative impacts on health,^{51–54} including both short-term and long-term impacts^{55,56} as well as physical and mental health impacts.^{43,57} Notably, conclusions regarding the importance of energy and climate actions can vary based on the health exposures and endpoints chosen for assessment. For instance, replacing solid fuels with electricity for cooking and heating reduces emissions of black carbon (BC), thereby lowering primary PM_{2.5} concentrations. In contrast, scaling up EVs can help avoid the NO_x emissions from gasoline and diesel combustion, which reduces the precursor emissions for both

PM_{2.5} and ozone. A wider consideration of health exposures and outcomes is therefore valuable in understanding the multi-dimensional health effects of climate-policy actions and the potential for co-benefits.

Second, future assessment will benefit from improved modeling capacity and uncertainty quantification relevant for finer scale assessment. For instance, the downscaling of emissions from state scale to fine scale plays a critical role in determining the potential emission hotspots in the future. Our main results (downscaled to 12 km × 12 km based on NEI 2017) and sensitivity analysis (downscaled to 1 km × 1 km for the industrial, building, and transportation sectors based on the Neighborhood Emission Mapping Operation [NEMO]⁵⁸; see Figure S12) demonstrates the importance of downscaling methods for some high-polluting counties and census tracts. Since some emissions hotspots, such as coal plants and coal-intensive facilities, will likely retire or shift to less-polluting technologies as part of the clean energy transition, future research needs to explore dynamic downscaling methods (i.e., considering the co-evolution of socioeconomic, population, energy, and emission activities over time), with careful treatment of relevant structural and parametric uncertainties.⁵⁹ In addition, energy system models such as GCAM are more capable of projecting energy-related emissions thanks to the detailed representation of technologies in the interactions between the power, industry, building, and transportation sectors. Future research needs to better represent the emissions from non-energy sectors, as emissions from wildfires and agriculture are expected to play a growing role in shaping air quality, particularly as energy-related emissions decline in a highly decarbonized world.⁶⁰

Furthermore, the model structure and complexity of the air-quality model also matters for the representation and characterization of fine-scale patterns. Although simpler models such as InMAP allow for rapid analysis of pollution impacts and the evaluation of numerous energy scenarios, they have limitations in terms of the pollutants they can simulate (e.g., ozone is often excluded due to its complex and non-linear chemistry) and the level of temporal detail they provide (e.g., annual rather than monthly or daily estimates). Additionally, they tend to perform less reliably when applied to different meteorological years^{61,62} or when facing significant shifts in emissions,^{59,63} both of which are important limitations of this study that warrant more careful investigation in future studies.

Third, while our study already makes an effort in this aspect, a more detailed representation of policy actors and their actions will still be useful to further improve the decision relevance of co-benefit assessment. This includes not only identifying different categories of policy actors (e.g., federal and state governments, companies, and households) but also various types of policy actions (e.g., subsidies, mandates, and tax).¹⁶ Representing these actors and actions in quantitative models raises important questions for model advancements. Despite growing attention on simulating policy heterogeneity in prior studies^{64,65} and the current analysis, operationalizing diverse policies in models remains difficult. Key methodological challenges include how to model and combine price-based versus quantitative-based policies, how to model dynamic changes over time such as technology learning, and how to represent potential policy interactions when one policy may make the other policy more (or less) effective.⁶⁶

Finally, a more thorough understanding of potential trade-offs is imperative. Our analysis focuses primarily on climate, pollution, and health goals where there is much room for synergy. However, equity is also an important policy consideration. Some studies suggest that designing energy decarbonization strategies based solely on least-cost principles may be less effective in mitigating exposure disparities compared to approaches that prioritize equity goals.^{30,67} This underscores potential equity-efficiency trade-offs that may arise from certain policy designs. In addition, a trade-off between different equity goals may also occur.^{42,67–69} For instance, policies that prohibit siting new polluting energy facilities close to disadvantaged communities could improve environmental justice issues related to pollution exposure but might worsen the inequities related to job and economic opportunities. While our analysis includes a sensitivity analysis focused on the distribution across counties with different socio-vulnerability index (Figure S11), future research should treat equity considerations in a more systematic way and carefully assess the potential trade-offs across regions, population groups, and policy goals.

METHODS

Scenario development

We consider a baseline scenario, i.e., the “baseline” scenario, as well as two climate-policy scenarios, i.e., the “existing climate policies” scenario and the “all-in climate policies” scenario. Both climate-policy scenarios include top-down and bottom-up policy representations, with the latter building on previously developed methodology for aggregating non-federal actions to the state level for implementation in GCAM-USA.^{1,70–73} We run the model from 2015 to 2030, for each of the 5-year time periods.

“Baseline” scenario

The “baseline” scenario considers the baseline changes in two main aspects: (1) the sociodemographic changes in 2030 compared to 2015, including changes in population, age structure, and improvements in general health conditions; and (2) business-as-usual air-pollution-control policies by assuming a declining share of polluting technologies, combined with technology- and fuel-specific emission factors of air pollutants with growing income level projected for the future. In addition, the scenario also considers future evolutions in technology costs based on the default reference case assumptions in GCAM-USA v.6 (see more in model codes and input files), such as declining cost of renewable technologies and electric vehicles. This scenario does not consider specific climate actions, such as carbon tax, clean energy mandates, or subsidies.

“Existing climate policies” scenario

The “existing climate policies” scenario includes existing federal policies—including several climate and energy provisions from the Bipartisan Infrastructure Law (BIL) and the IRA—and non-federal policies such as regional CO₂ emission caps, state-level renewable energy mandates, and state-level electric vehicle incentives. Altogether, we find that these existing policies can reduce GHG emissions by 40% below 2005 levels by 2030. Existing state-level renewable portfolio standard (RPS), corporate average fuel economy (CAFE) standards, and several tax credits from the IRA for renewable electricity generation, zero-emissions

vehicles (ZEVs), and carbon capture and storage (CCS) are among the key policy drivers in this scenario.

“All-in climate policies” scenario

The IRA, along with additional policies and actions from Congress, the federal government, states, cities, and businesses, collectively provide a major boost to climate action in the US. Yet these existing policies on their own will not be enough for the US to meet its 2030 climate target and achieve ambitious decarbonization through 203 and beyond. Thus, our “all-in climate policies” scenario models GHG emission reductions achievable under a comprehensive, all-in climate strategy that builds upon the policy framework in the “existing climate policies” scenario with enhanced non-federal and federal action. Some of the additional climate actions modeled in this scenario include a full phaseout of unabated coal-fired electricity generation by 2030, accelerated adoption of ZEVs in light-duty vehicle, bus, and freight truck markets, ramped-up electric appliance standards, more stringent standards, and oil and gas methane, as well as extensions of the existing IRA tax credits beyond their legislated sunset dates. Altogether, these actions have the potential to deliver a 52% reduction in GHG emissions from 2005 levels by 2030, achieving the 2030 US climate target.

Overview of GCAM-USA-CGS

Our analysis uses a version of the open-source GCAM to estimate the aggregate impact of federal and non-federal climate policies and actions on economy-wide emissions reductions in the US, including both GHG and criteria air pollutants. Specifically, we use GCAM-USA, a state-level version of GCAM. We refer to the version of GCAM-USA used in this study as GCAM-USA-CGS (Center for Global Sustainability).

GCAM is an integrated assessment model that covers the energy, land, water, climate, and socioeconomic systems. The global version of GCAM groups the world’s countries into 32 geopolitical regions with representation of the energy and socioeconomic systems for each region. The US is one of the 32 regions. GCAM represents the global climate system and uses 235 water basins and 384 land regions to represent global water and land systems. GCAM tracks emissions and sinks of carbon dioxide (CO₂), 16 other GHGs, and several air pollutants.

GCAM is a market equilibrium model. The model solves for equilibrium in each period by finding a set of market prices such that supplies and demands are equal in all markets as the actors in the model adjust the quantities of the commodities they buy and sell. GCAM operates in 5-year time increments, with each new period starting from the conditions that emerged in the last, with most technologies vintaged such that a portion of existing stocks at any point carry over into future time periods.

The state-level version of GCAM used in this analysis, GCAM-USA, disaggregates the US energy and economy components into 50 states and the District of Columbia while maintaining the same level of detail as GCAM for water and land sectors. The energy system in GCAM-USA has representations of depletable primary sources such as coal, gas, oil, and uranium, in addition to renewable resources such as biomass, hydropower, solar, wind, and geothermal. Energy transformation

processes such as oil refining and electricity generation are represented at the state level in GCAM-USA. These energy carriers, in turn, are used to deliver services to state-level end users in the buildings, transportation, and industrial sectors. The electric power sector includes representations of a range of power generation technologies, including those fueled by fossil fuels and bioenergy (with and without CCS), renewables, and nuclear power.

GCAM-USA-CGS is based on the open-source release of GCAM-USA 6.0.⁷⁴ GCAM-USA-CGS has been updated for the purposes of this study to reflect changes such as the most recent estimates of future renewable energy costs.⁷⁵ The model is also calibrated to the latest non-CO₂ marginal abatement cost curves from the US Environmental Protection Agency (EPA).⁷⁶

The model reports the annual emissions of both CO₂ and major criteria air pollutants, including SO₂, NO_x, NH₃, volatile organic compounds, primary PM_{2.5}, BC, and organic carbon (OC). The air-pollutant emissions from the power, industry, building, and transportation sectors are estimated at the state level, whereas the emissions from land use and other agricultural sectors are reported at the national level.

Emission mapping and downscaling

We use a proportional downscaling approach to downscale the GCAM-USA-CGS state-level emissions to 12-km by 12-km spatial grids. For the emissions from the agricultural, industrial, power, transportation, and building sectors, we assume that the emissions follow the spatial and vertical pattern of the 2017 emissions data from the NEI database.³⁸ For the other land-use sectors (primarily from fire), we assume that the emissions are the same as “baseline” scenario and downscale the emissions following the patterns of the 2017 Fire Inventory from the National Center for Atmospheric Research (FINN).⁷⁷

Specifically, for each air pollutant from each aggregate sector, we first sum up the hourly emissions from NEI and FINN 2017 to calculate the annual total emissions for each 12-km × 12-km spatial grid as well as the state total. We then calculate the share of each grid in-state-total emissions as the downscaling factor. Based on the state-level emissions from GCAM-USA-CGS and the downscaling factors, we obtain the gridded emissions by sector and by pollutant type.

We further assume the downscaling factors, which reflect the spatial distribution within each state, remain the same from 2015 to 2030. For the air-pollutant emissions outside the continental US (e.g., Canada and Mexico), we keep their emissions at the 2017 NEI and FINN levels to tease out the effects of domestic emissions.

Since fine-scale emission patterns largely affect the fine-scale pollution patterns,^{78,79} we consider an alternative, hybrid downscaling approach whereby we use NEI to downscale power sector emissions to 12-km × 12-km resolution and use the NEMO database to downscale the emissions from building, transportation, and non-point sources in the industrial sector to 1-km × 1-km resolution.⁵⁸ As shown in Figure S12, we find this hybrid downscaling method to result in similar air-pollution patterns in most regions compared to our main results. Yet we find different high-pollution locations to emerge in some regions, such as California, which suggests that the downscaling approach matters for fine-scale pollution hotspots.

Air-quality modeling

For each scenario and model year, we use the gridded emissions to simulate the annual average ambient concentrations of PM_{2.5} using a reduced-complexity, national-scale air-quality model, InMAP.³⁹ This model simulates the primary ambient PM_{2.5} and secondary ambient PM_{2.5} (including SO₄²⁻, NO₃⁻, and NH₄⁺) from the emissions from SO₂, NO_x, NH₃, volatile organic compounds, and primary PM_{2.5} emissions. For most emissions, we use those projected and downscaled from GCAM-USA-CGS. For primary PM_{2.5} emissions from the agricultural sector, we use the following equation to convert GCAM-reported BC and OC emissions to primary PM_{2.5}.^{80,81}

$$\text{Primary PM}_{2.5} = (\text{BC} + \text{OC} * 1.8) * 1.1 \quad (\text{Equation 1})$$

InMAP is operated at various spatial resolutions, ranging from 1 km × 1 km in urban centers to 48 km × 48 km in areas with low population density. It simulates 29 vertical layers with a ground-level height of roughly 30 m. InMAP simulates ambient pollutant concentrations by estimating a steady-state solution to the reaction-advection-diffusion equation.³⁹ In this equation, the advection term accounts for the transport of pollutants by the wind. It captures the movement of pollutants along with the air flow, based on wind speed and direction. The diffusion term models the dispersion of pollutants in the atmosphere due to turbulence and molecular mixing. Diffusion spreads pollutants from areas of higher concentration to areas of lower concentration, both horizontally and vertically. The reaction term captures the chemical transformations that pollutants undergo in the atmosphere, such as how primary pollutants form secondary particles like PM_{2.5}. The exogenous parameters include wind speed and direction, diffusion and convection coefficients, temperature, and chemical reaction coefficients. They are preprocessed using the Weather Research and Forecasting model coupled with chemistry (WRF-Chem) simulations, which were conducted for each of the over 50,000 grid cells covering the continental US using the 2005 meteorological field.³⁹ InMAP assumes a steady-state condition, meaning that pollutant concentrations do not change significantly over short timescales within the grid. This simplifies the equation by assuming that the rates of emission, advection, diffusion, and reaction balance each other out over time, allowing the model to solve for average concentrations rather than simulating time-varying dynamics.

The InMAP model performance has been evaluated in previous studies.³⁹ Broadly speaking, InMAP performs well in capturing the general spatial and temporal patterns, though it often underestimates in highly polluting areas.³⁹ It also has a less satisfying performance when applied to a different meteorological year (see Figure S13 for air-quality evaluation results using WRF-Chem and InMAP). Therefore, although InMAP is not viewed as the best tool for air-quality assessment, as a computationally efficient tool, InMAP has been used widely to couple with energy modeling to provide a rough estimate of the air-quality impacts of energy transition.^{44,61}

Health-impact assessment

We assess premature deaths associated with long-term exposure to ambient PM_{2.5} in each of 3,108 counties across the continental US using BenMAP¹² version 1.5.8. BenMAP is an

Table 3. Summary of input data for the health-impact assessment

Variable	Definition	Data source
Y_0	all-cause baseline mortality rate	for 2015: Center for Disease Control (CDC) WONDER database for 2020–2030: projections from BenMAP ¹²
Pop	county-level population	for 2015: projections from 2010 US Census Bureau census block data ³² for 2020–2030: projections using time-series models ARIMA from the Socioeconomic Data and Application Center ^{33,34}
β	changes in mortality risk resulting from changes in $PM_{2.5}$ exposure level	here we use the estimate from the American Cancer Society ⁴⁵ ; we use the same estimate each year from 2015 to 2030
ΔPM	county-level $PM_{2.5}$ concentration	county-level $PM_{2.5}$ concentrations averaged from gridded concentrations simulated by InMAP; we use the simulated results from 2015 to 2030

open-source computer program developed by the EPA. It estimates the number of premature deaths associated with long-term air-pollution exposure as one of the health metrics. BenMAP-CE utilizes a comprehensive database that includes concentration-response relationships and demographic, health, and economic data to quantify these impacts, supporting users in assessing the health benefits of improved air quality.

To estimate the county-level exposure to ambient $PM_{2.5}$, we use InMAP-simulated $PM_{2.5}$ concentrations at varying resolutions from 1 km × 1 km to 48 km × 48 km: if the InMAP grid size is smaller than a county, we calculate a population-weighted average $PM_{2.5}$ concentration for the county using multiple InMAP grids. If the InMAP grid size is larger than a county, we use the same $PM_{2.5}$ concentrations for all the counties within one InMAP grid.

Our main results use the following Krewski et al. log-linear health-impact function⁴⁵ from BenMAP to calculate all-cause deaths attributable to long-term ambient $PM_{2.5}$ exposure. The detailed information for each component is presented in Table 3.

$$\Delta Y = (1 - e^{\beta \cdot \Delta PM}) \cdot Y_0 \cdot \text{Pop} \quad (\text{Equation 2})$$

As a sensitivity analysis, we also consider an alternative exposure-response function based on the GEMM¹³ (Figure S14). Consistent with the literature,¹³ projected $PM_{2.5}$ -attributable deaths results using GEMM are around 200% higher than the results using the Krewski et al. log-linear function.⁴⁵ However, the relative differences across scenarios, as well as the spatial distribution of $PM_{2.5}$ -attributable deaths across subnational regions, remain similar.

Decomposition analysis

Based on prior studies,^{11,85} four key factors influence the total mortality attributable to ambient $PM_{2.5}$, namely (1) population growth, (2) aging, (3) baseline mortality rates independent of $PM_{2.5}$ exposure, and (4) the levels of exposure to ambient $PM_{2.5}$. In this decomposition analysis, we quantify the individual contributions of each factor $PM_{2.5}$ -attributable deaths, in the sequence from (1) to (4). We focus on the changes in the “baseline” scenario and two climate-policy scenarios in 2030 relative to 2015 and quantify the effects of changing each factor at a time.^{11,86} Factor (4) is divided to two parts—changes in the levels of exposure to ambient $PM_{2.5}$ due to (i) business-as-usual air-pollution-control efforts and (ii) climate policies. In the “baseline” scenario, only part (i) is considered; in the two climate-policy

scenarios, both parts are included. We perform this analysis at the national level.

Specifically, we first estimate the $PM_{2.5}$ -attributable mortality in 2015 and 2030 in each scenario:

$$^{2015} \text{Mortality} = \sum_{c,a} (^{2015} Y_0)_{c,a} \cdot ^{2015} \text{Pop}_{c,a} \cdot ^{2015} \text{AF}(\Delta PM) \quad (\text{Equation 3})$$

$$^{2030} \text{Mortality} = \sum_{c,a} (^{2030} Y_0)_{c,a} \cdot ^{2030} \text{Pop}_{c,a} \cdot ^{2030} \text{AF}(\Delta PM) \quad (\text{Equation 4})$$

where c represents one of each county, and a represents one of each age group. $\text{AF} = 1 - e^{\beta \cdot \Delta PM}$ is the attributable fraction and is a function of the $PM_{2.5}$ concentration.

We then define:

$$A = ^{2015} \text{Mortality} \cdot \frac{^{2030} \text{Pop}_{c,a}}{^{2015} \text{Pop}_{c,a}} \quad (\text{Equation 5})$$

$$B = \sum_{c,a} (^{2015} Y_0)_{c,a} \cdot ^{2030} \text{Pop}_{c,a} \cdot ^{2015} \text{AF}(\Delta PM) \quad (\text{Equation 6})$$

$$C = \sum_{c,a} (^{2030} Y_0)_{c,a} \cdot ^{2030} \text{Pop}_{c,a} \cdot ^{2015} \text{AF}(\Delta PM) \quad (\text{Equation 7})$$

Using the definitions above, we have the following.

Effects from sociodemographic changes:

Population growth effect (%) = $(A - \text{mortality in 2015}) / \text{mortality in 2015}$

Population aging effect (%) = $(B - A) / \text{mortality in 2015}$

Baseline mortality rate change effect (%) = $(C - B) / \text{mortality in 2015}$

Effects from changes in $PM_{2.5}$ exposure:

Exposure change effect (%) from business-as-usual air-pollution-control efforts = $(\text{mortality in “baseline” 2030} - C) / \text{mortality in 2015}$

Exposure change effect (%) from climate policies = $(\text{mortality in climate-policy scenarios in 2030} - \text{mortality in “baseline” 2030}) / \text{mortality in 2015}$

Table 4. Alternative assumptions to assess policy coordination and enforcement

Policy consideration	Main “all-in climate policies” scenario	Sensitivity analysis
Imperfect enforcement of climate policy	narrative: perfect enforcement of federal, state, and local climate and energy policies model assumption: energy activities and associated emissions are projected using GCAM-USA-CGS, assuming all climate and clean energy actions are perfectly enforced	narrative: considered climate and energy policies are moderately enforced model assumption: compared to the reductions in air-pollutant emissions in the main “all-in climate policies” scenario to the “baseline” scenario, here we assume the reduction is 20% less due to imperfect enforcement
More stringent pollution-control efforts	narrative: the air-pollutant standards are tightened gradually over time based on historical patterns ⁵⁷ model assumption: trends for emissions factor changes are derived from historical data ⁸⁸ and are adjusted for the future based on a declining trend with increasing income ³⁴	narrative: more stringent air-pollution-control regulations and standards are adopted at federal, state, and local scales model assumption: emission factors of fossil fuels in the power, industry, building, and transportation sectors are reduced by 20% compared to the main scenario
Greater improvement in general health conditions	narrative: moderate improvement in general health conditions over time model assumption: PM _{2.5} -attributable baseline mortality rates are set at the SSP2 level, which is a middle-of-the-road scenario	narrative: greater improvement in general health conditions model assumption: PM _{2.5} -attributable baseline mortality rates are set at the SSP1 level, which assumes more advanced socioeconomic development trends.
Less improvement in general health conditions	narrative: same as above model assumption: same as above	narrative: less improvement in general health conditions model assumption: PM _{2.5} -attributable baseline mortality rates are increased to the SSP3 level, which assumes limited socioeconomic development trends

Total effect:

Total change (%) = (mortality in 2030 – mortality in 2015)/mortality in 2015

Sensitivity analysis of policy coordination and enforcement

With a focus on the “all-in climate policies” scenario, we design sensitivity analysis to evaluate the effects of three factors related to policy coordination and enforcement: (1) imperfect enforcement of climate policy; (2) more stringent air-pollution-control efforts; and (3) faster or slower improvements in general health conditions. These assumptions are summarized in Table 4.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Dr. Wei Peng (weipeng@princeton.edu).

Materials availability

The study did not generate new unique materials.

Data and code availability

All original code and data used in the current draft are available from <https://doi.org/10.5281/zenodo.10830268>.

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AUTHOR CONTRIBUTIONS

X.H., A.Z., and W.P. conceived and designed the analysis. X.H. performed the air quality and health analysis. A.Z. performed all GHG analysis. A.Z., Y.O., S.K., G.I., H.M., R.C., and N.H. developed the mitigation scenario and contributed to the modeling tool. X.H. and W.P. led the writing of the manuscript with input from all authors. All authors reviewed the data, provided feedback, and contributed to the final manuscript.

DECLARATION OF INTERESTS

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SUPPLEMENTAL INFORMATION

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One Earth, Volume 8

Supplemental information

**Substantial air quality and health co-benefits
from combined federal and subnational climate
actions in the United States**

Xinyuan Huang, Wei Peng, Alicia Zhao, Yang Ou, Shannon Kennedy, Gokul Iyer, Haewon McJeon, Ryna Cui, and Nate Hultman

Table S1. Definition of terms and examples

Key term	Definition	Example
Health benefits	Health benefits refer to the direct, intended improvements in public health because of specific actions or policies. These are the primary goals of health-related regulations or interventions.	Reducing air pollution through stricter emissions standards directly leads to fewer respiratory diseases and improved public health outcomes.
Health co-benefits	Health co-benefits are additional benefits related to the reduction of greenhouse gas emissions that are not directly related to climate change, such as air quality improvement, technological innovation, or employment creation. ¹	Health co-benefits are more strictly defined compared to health benefits. For example, reduction in PM _{2.5} -attributable deaths due to the implementation of All-in climate actions are health co-benefits.
Health equity	Health equity is the state in which everyone has a fair and just opportunity to attain their highest level of health ² , regardless of their racial or ethnic background, age, or geographic location.	Health equity is improved when regions that currently suffer from higher health risks benefit more from climate actions
Climate policy	Climate policy encompasses the regulations, strategies, and actions designed to mitigate the impacts of climate change and adapt to its effects. These policies	Biden administration setting a goal of reducing carbon emissions by 50% to 52% compared to 2005 levels by 2030, a carbon free power sector by 2035, and for the

	typically focus on reducing greenhouse gas emissions, promoting renewable energy, and enhancing climate resilience. ³	entire economy to be net zero by 2050.
Air pollution control efforts	Air pollution control efforts are the regulatory frameworks aimed at controlling emissions of pollutants that harm public health and the environment. These efforts set limits on pollutants like particulate matter, nitrogen oxides, and sulfur dioxide, often through emissions standards for industries and vehicles. Air pollution control efforts can also be linked to climate and energy policies, as reducing fossil fuel use often leads to lower emissions of harmful air pollutants. ⁴	The National Ambient Air Quality Standards (NAAQS) are limits set by the Environmental Protection Agency (EPA) for certain pollutants in outdoor air, intended to protect public health and the environment. These standards are part of the Clean Air Act and focus on six key pollutants, known as criteria pollutants: Particulate matters, ozone, carbon monoxide, nitrogen dioxide, sulfur dioxide, and lead.

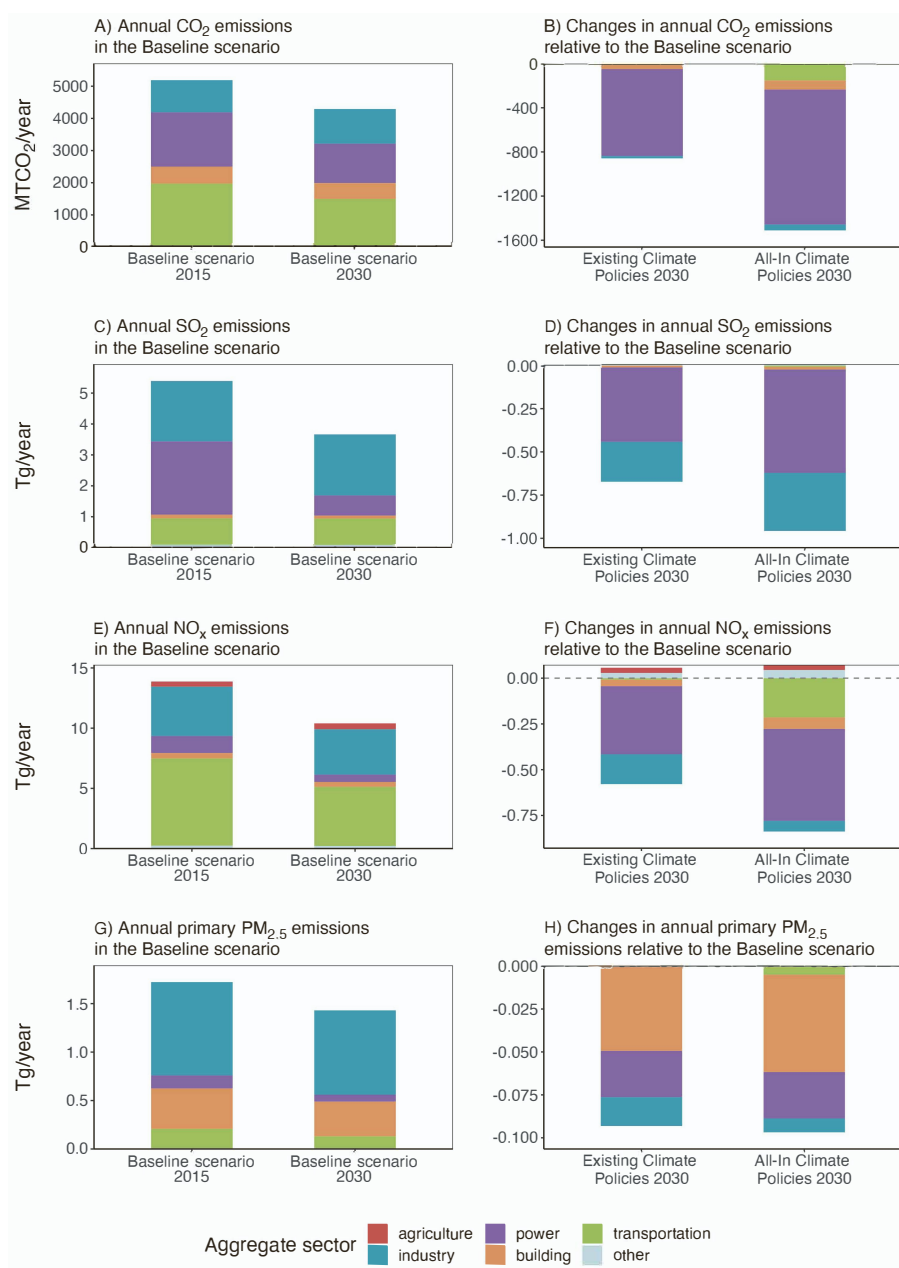


Figure S1. Changes of national CO₂ and air pollutants emissions by sector. Panel A) shows the annual CO₂ emissions in the Baseline scenario in 2015 and 2030. Panel B) shows the changes in annual CO₂ emissions in the two policy scenarios relative to the “Baseline” scenario in 2030. Panel C) shows the annual SO₂ emissions in the Baseline scenario in 2015 and 2030. Panel D) shows the changes in annual SO₂ emissions in the two policy scenarios relative to the Baseline scenario in 2030. Panel E) shows the annual NO_x emissions in the Baseline scenario in 2015 and 2030. Panel F) shows the changes in annual NO_x emissions in the two policy scenarios relative to the Baseline scenario in 2030. Panel G) shows the annual

primary PM_{2.5} emissions in the Baseline scenario in 2015 and 2030. Panel H) shows the changes in annual primary PM_{2.5} emissions in the two policy scenarios relative to the Baseline scenario in 2030. Here the “Other” emissions consider land use activities projected by the GCAM-USA model. Here we focus primarily on energy-related emissions and do not consider fire emissions that will likely become a major pollution source in the future.

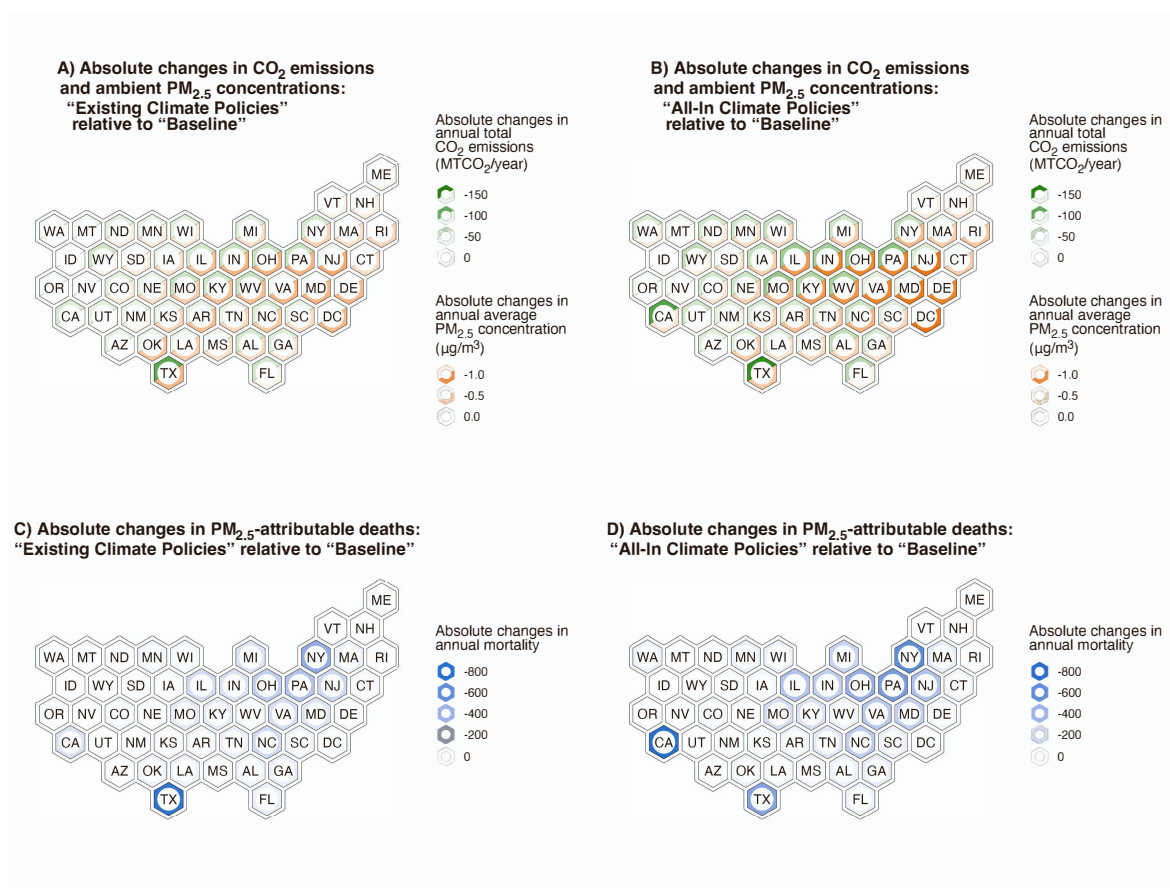


Figure S2. State-level absolute reductions in CO₂ emissions, PM_{2.5} exposure, and PM_{2.5}-attributable deaths in 2030. Panels A) and B) show the state-level absolute reductions in CO₂ emissions and ambient PM_{2.5} concentrations in the "Existing Climate Policies" scenario and the "All-In Climate Policies" scenario compared to the "Baseline" scenario, respectively. Panels C) and D) show the state-level absolute reductions in PM_{2.5}-attributable health burden in the "Existing Climate Policies" scenario and the "All-In Climate Policies" scenario compared to the "Baseline" scenario, respectively.

Table S2. Relative and absolute reductions of PM_{2.5}-attributable deaths in all states in 2030.

State	Population	Baseline scenario	Existing Climate Policies	Δ%	All-In Climate Policies	Δ%
Alabama	5096509	1001	941.5	-5.9%	922.9	-7.8%
Arizona	8181257	618	608.9	-1.4%	602.8	-2.4%
Arkansas	3200915	695	636.5	-8.4%	634.9	-8.7%
California	44504303	10636	10481.9	-1.4%	9848.5	-7.4%
Colorado	6879984	683	643.7	-5.7%	629.2	-7.8%
Connecticut	3606564	386	350.0	-9.3%	333.9	-13.5%
Delaware	1077980	284	259.3	-8.8%	241.1	-15.2%
District of Columbia	899517	151	139.8	-7.3%	129.4	-14.2%
Florida	24873435	3059	2976.0	-2.7%	2985.4	-2.4%
Georgia	11944103	1809	1725.3	-4.6%	1712.8	-5.3%
Idaho	1947380	72	71.9	-0.5%	71.0	-1.9%
Illinois	12835693	2654	2471.7	-6.9%	2339.3	-11.8%
Indiana	7045980	1677	1517.7	-9.5%	1414.5	-15.6%
Iowa	3410253	586	558.3	-4.7%	541.9	-7.5%
Kansas	3102548	716	683.1	-4.5%	676.9	-5.4%
Kentucky	4690750	821	721.4	-12.2%	663.4	-19.2%
Louisiana	5153855	1357	1288.1	-5.1%	1322.5	-2.5%
Maine	1320925	68	62.3	-8.7%	60.0	-12.0%
Maryland	6686651	1397	1258.3	-9.9%	1146.6	-17.9%
Massachusetts	7477860	742	671.1	-9.6%	638.2	-14.0%
Michigan	10102384	1889	1744.1	-7.7%	1659.6	-12.1%
Minnesota	6090474	812	759.7	-6.5%	738.2	-9.1%
Mississippi	3086444	506	473.7	-6.3%	470.3	-7.0%
Missouri	6366147	1368	1254.7	-8.3%	1188.0	-13.2%
Montana	1174972	56	55.8	-0.7%	54.8	-2.4%
Nebraska	2143005	291	274.0	-5.7%	266.6	-8.3%
Nevada	3541326	680	676.1	-0.6%	668.5	-1.7%
New Hampshire	1352067	96	86.9	-9.7%	83.0	-13.7%
New Jersey	9427373	2434	2228.3	-8.5%	2098.3	-13.8%
New Mexico	2127668	146	139.3	-4.7%	138.2	-5.4%
New York	20881206	5255	4805.6	-8.5%	4634.8	-11.8%
North Carolina	11632866	2235	2039.7	-8.7%	1960.5	-12.3%
North Dakota	1066340	47	43.8	-7.7%	41.9	-11.8%
Ohio	11870016	2547	2284.7	-10.3%	2118.2	-16.9%
Oklahoma	4487827	1023	951.0	-7.0%	955.7	-6.6%
Oregon	4681828	310	301.7	-2.7%	297.0	-4.2%
Pennsylvania	13043383	3115	2794.9	-10.3%	2571.1	-17.5%
Rhode Island	1068021	128	114.7	-10.5%	109.2	-14.8%
South Carolina	5759261	1055	990.6	-6.1%	974.6	-7.6%

South Dakota	1130550	114	106.9	-6.0%	103.3	-9.2%
Tennessee	7415821	1060	973.7	-8.1%	939.0	-11.4%
Texas	35329058	9504	8735.4	-8.1%	9010.3	-5.2%
Utah	3784016	126	123.0	-2.6%	121.4	-4.0%
Vermont	612305	34	30.7	-8.6%	29.6	-12.0%
Virginia	9485611	1728	1558.0	-9.9%	1434.6	-17.0%
Washington	8643356	800	782.4	-2.2%	727.3	-9.1%
West Virginia	1792058	311	271.2	-12.8%	244.5	-21.4%
Wisconsin	6016174	897	848.6	-5.4%	821.4	-8.4%
Wyoming	668313	45	43.2	-4.9%	42.2	-7.1%

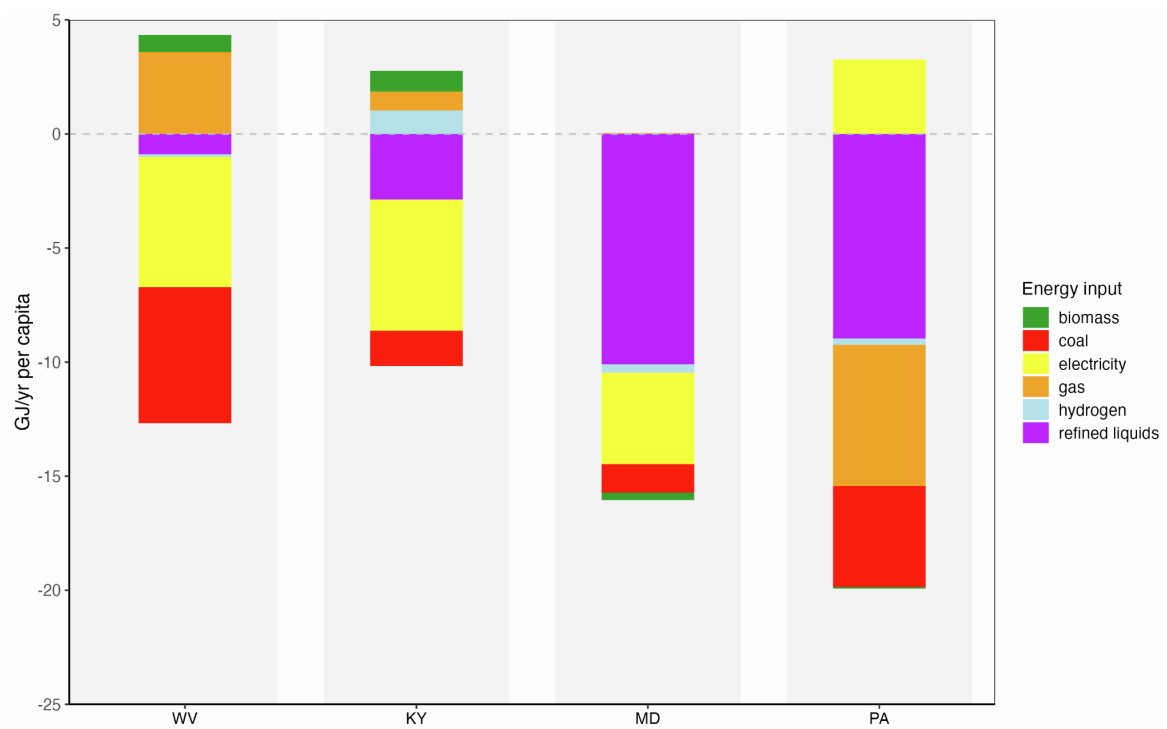


Figure S3. Changes in final energy use in selected Mid-Atlantic states, in the “All-In Climate Policies” scenario relative to the “Baseline” scenario, in 2030. Positive values indicate increase in this energy input, and negative values indicate decrease in this energy input.

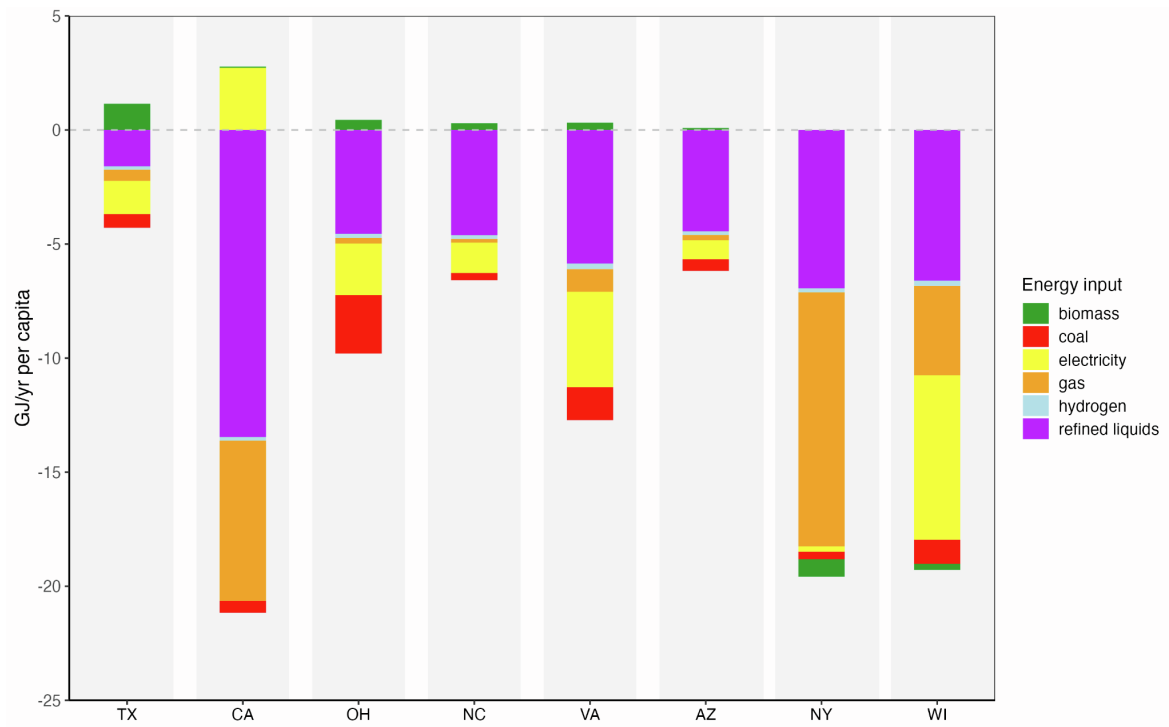


Figure S4. Changes in final energy use in selected states, in the “All-In Climate Policies” scenario relative to the Baseline scenario, in 2030. Positive values indicate increases in this energy input, and negative values indicate decreases in this energy input.

Table S3. State-level changes in final energy use and CO₂ emissions in the policy scenarios relative to the Baseline scenario in 2030.

State	Energy	Baseline		Existing Climate Policies		All-In Climate Policies	
		Input (EJ)	CO ₂ (10 ⁶ Ton)	Input (EJ)	CO ₂ (10 ⁶ Ton)	Input (EJ)	CO ₂ (10 ⁶ Ton)
AL	biomass	0.16	99.5	0.17	70.5	0.17	58.0
	coal	0.08		0.07		0.05	
	electricity	0.39		0.39		0.38	
	gas	0.24		0.23		0.24	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.44		0.44		0.42	
AR	biomass	0.07	49.3	0.07	32.5	0.08	23.9
	coal	0.01		0.01		0.01	
	electricity	0.20		0.20		0.19	
	gas	0.15		0.14		0.15	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.23		0.24		0.22	
AZ	biomass	0.00	60.9	0.00	45.1	0.00	38.4
	coal	0.01		0.01		0.01	
	electricity	0.36		0.35		0.35	
	gas	0.08		0.07		0.08	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.43		0.44		0.40	
CA	biomass	0.06	340.6	0.06	316.1	0.07	227.0
	coal	0.06		0.06		0.04	
	electricity	1.31		1.37		1.42	
	gas	1.49		1.41		1.22	
	hydrogen	0.02		0.03		0.01	
	refined liquids	3.11		3.03		2.59	
CO	biomass	0.01	72.5	0.01	52.0	0.01	43.0
	coal	0.01		0.01		0.01	
	electricity	0.27		0.25		0.27	
	gas	0.35		0.32		0.31	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.39		0.38		0.34	
CT	biomass	0.01	25.9	0.01	22.3	0.01	18.9
	coal	0.00		0.00		0.00	
	electricity	0.13		0.14		0.14	

	gas	0.12		0.11		0.10	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.23		0.21		0.19	
DC	biomass	0.00	2.5	0.00	2.3	0.00	1.8
	coal	0.00		0.00		0.00	
	electricity	0.05		0.04		0.04	
	gas	0.03		0.02		0.02	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.03		0.03		0.03	
DE	biomass	0.00	9.7	0.00	8.6	0.00	7.0
	coal	0.00		0.00		0.00	
	electricity	0.04		0.04		0.04	
	gas	0.04		0.04		0.04	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.06		0.06		0.05	
FL	biomass	0.13	195.0	0.13	163.7	0.13	145.9
	coal	0.03		0.03		0.02	
	electricity	1.00		0.96		0.94	
	gas	0.29		0.30		0.31	
	hydrogen	0.01		0.01		0.01	
	refined liquids	1.44		1.46		1.40	
GA	biomass	0.21	102.6	0.21	81.5	0.22	73.9
	coal	0.01		0.01		0.01	
	electricity	0.59		0.57		0.56	
	gas	0.30		0.29		0.30	
	hydrogen	0.01		0.01		0.00	
	refined liquids	0.68		0.70		0.66	
IA	biomass	0.02	36.3	0.02	22.9	0.02	-4.6
	coal	0.05		0.05		0.04	
	electricity	0.20		0.20		0.20	
	gas	0.19		0.19		0.19	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.29		0.29		0.27	
ID	biomass	0.03	13.8	0.03	12.6	0.03	11.6
	coal	0.00		0.00		0.00	
	electricity	0.11		0.11		0.11	
	gas	0.06		0.06		0.06	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.16		0.16		0.15	

IL	biomass	0.01	154.0	0.01	123.8	0.02	87.4
	coal	0.08		0.07		0.06	
	electricity	0.63		0.63		0.62	
	gas	0.71		0.66		0.59	
	hydrogen	0.01		0.01		0.01	
	refined liquids	0.96		0.96		0.84	
IN	biomass	0.03	140.0	0.03	104.0	0.03	77.7
	coal	0.16		0.14		0.12	
	electricity	0.46		0.46		0.43	
	gas	0.48		0.47		0.49	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.58		0.59		0.57	
KS	biomass	0.01	50.5	0.01	33.2	0.01	26.2
	coal	0.01		0.01		0.01	
	electricity	0.17		0.17		0.17	
	gas	0.20		0.19		0.19	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.25		0.25		0.25	
KY	biomass	0.03	101.9	0.03	67.9	0.03	48.1
	coal	0.03		0.02		0.02	
	electricity	0.33		0.32		0.31	
	gas	0.17		0.16		0.17	
	hydrogen	0.00		0.01		0.01	
	refined liquids	0.44		0.44		0.43	
LA	biomass	0.12	184.5	0.12	171.1	0.14	165.2
	coal	0.00		0.00		0.00	
	electricity	0.35		0.35		0.33	
	gas	1.29		1.30		1.30	
	hydrogen	0.00		0.00		0.00	
	refined liquids	2.02		2.01		2.04	
MA	biomass	0.02	48.1	0.02	42.2	0.02	35.9
	coal	0.00		0.00		0.00	
	electricity	0.25		0.27		0.26	
	gas	0.26		0.24		0.20	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.44		0.41		0.37	
MD	biomass	0.01	45.4	0.01	38.3	0.01	29.6
	coal	0.03		0.04		0.02	
	electricity	0.27		0.25		0.24	

	gas	0.14		0.14		0.14	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.38		0.35		0.32	
ME	biomass	0.09	12.8	0.08	11.6	0.08	9.4
	coal	0.00		0.00		0.00	
	electricity	0.05		0.06		0.06	
	gas	0.03		0.03		0.03	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.15		0.15		0.13	
MI	biomass	0.09	114.5	0.09	95.4	0.09	80.4
	coal	0.07		0.06		0.05	
	electricity	0.49		0.47		0.46	
	gas	0.58		0.54		0.56	
	hydrogen	0.01		0.01		0.00	
	refined liquids	0.66		0.68		0.62	
MN	biomass	0.05	60.8	0.05	46.9	0.05	29.3
	coal	0.02		0.01		0.01	
	electricity	0.29		0.29		0.28	
	gas	0.28		0.26		0.24	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.47		0.47		0.42	
MO	biomass	0.02	117.5	0.02	80.7	0.02	48.9
	coal	0.04		0.04		0.03	
	electricity	0.36		0.36		0.36	
	gas	0.24		0.24		0.24	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.48		0.49		0.45	
MS	biomass	0.05	52.2	0.05	43.3	0.06	39.2
	coal	0.00		0.00		0.00	
	electricity	0.21		0.21		0.20	
	gas	0.14		0.13		0.14	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.35		0.35		0.33	
MT	biomass	0.01	24.3	0.01	15.4	0.01	12.3
	coal	0.01		0.01		0.00	
	electricity	0.06		0.06		0.06	
	gas	0.05		0.04		0.05	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.12		0.12		0.12	

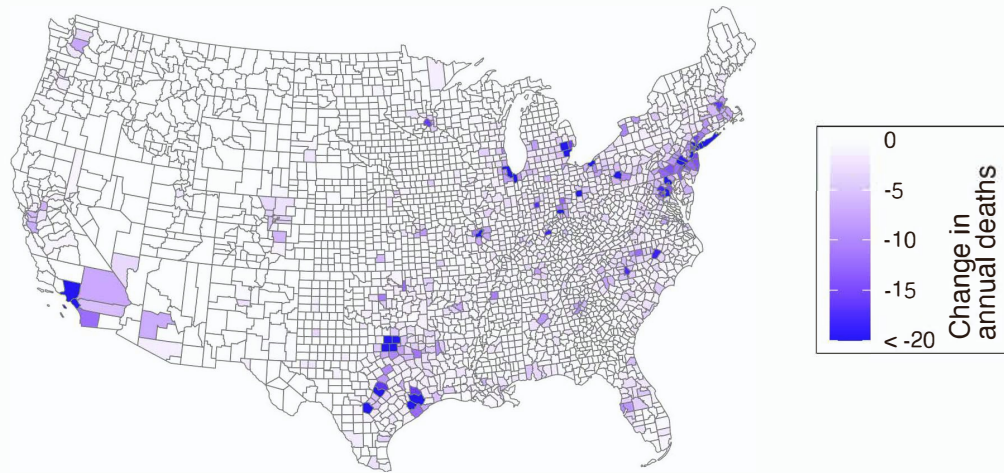
NC	biomass	0.09	93.9	0.09	68.0	0.09	58.5
	coal	0.01		0.01		0.01	
	electricity	0.57		0.55		0.55	
	gas	0.22		0.21		0.21	
	hydrogen	0.01		0.01		0.00	
	refined liquids	0.62		0.62		0.57	
ND	biomass	0.00	47.1	0.00	27.8	0.00	15.7
	coal	0.08		0.07		0.06	
	electricity	0.07		0.08		0.07	
	gas	0.06		0.06		0.06	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.13		0.13		0.13	
NE	biomass	0.00	33.6	0.00	18.2	0.00	0.0
	coal	0.02		0.02		0.02	
	electricity	0.12		0.12		0.12	
	gas	0.08		0.08		0.08	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.18		0.18		0.17	
NH	biomass	0.02	10.4	0.01	9.0	0.01	6.9
	coal	0.00		0.00		0.00	
	electricity	0.05		0.05		0.05	
	gas	0.02		0.02		0.02	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.11		0.11		0.09	
NJ	biomass	0.01	94.6	0.01	84.4	0.01	69.3
	coal	0.00		0.00		0.00	
	electricity	0.35		0.35		0.35	
	gas	0.40		0.36		0.30	
	hydrogen	0.00		0.01		0.01	
	refined liquids	0.86		0.82		0.74	
NM	biomass	0.01	41.7	0.01	26.6	0.01	21.9
	coal	0.00		0.00		0.00	
	electricity	0.11		0.12		0.11	
	gas	0.15		0.15		0.14	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.19		0.18		0.16	
NV	biomass	0.00	26.4	0.00	22.6	0.00	20.1
	coal	0.01		0.01		0.01	
	electricity	0.16		0.17		0.17	

	gas	0.08		0.07		0.07	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.19		0.19		0.18	
NY	biomass	0.07	142.7	0.06	114.7	0.06	101.7
	coal	0.02		0.02		0.02	
	electricity	0.71		0.70		0.70	
	gas	0.83		0.71		0.61	
	hydrogen	0.01		0.01		0.00	
	refined liquids	1.19		1.14		1.06	
OH	biomass	0.05	169.6	0.05	131.7	0.05	103.6
	coal	0.12		0.10		0.09	
	electricity	0.67		0.66		0.64	
	gas	0.67		0.65		0.66	
	hydrogen	0.01		0.01		0.00	
	refined liquids	0.86		0.87		0.81	
OK	biomass	0.03	69.7	0.03	57.1	0.03	48.7
	coal	0.01		0.01		0.01	
	electricity	0.26		0.26		0.26	
	gas	0.34		0.33		0.33	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.39		0.39		0.37	
OR	biomass	0.06	30.0	0.06	25.8	0.06	22.6
	coal	0.00		0.00		0.00	
	electricity	0.22		0.23		0.22	
	gas	0.10		0.10		0.09	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.28		0.27		0.25	
PA	biomass	0.09	180.5	0.09	142.7	0.09	105.2
	coal	0.20		0.18		0.15	
	electricity	0.65		0.65		0.69	
	gas	0.71		0.69		0.63	
	hydrogen	0.01		0.01		0.00	
	refined liquids	0.92		0.92		0.80	
RI	biomass	0.00	7.9	0.00	6.6	0.00	5.5
	coal	0.00		0.00		0.00	
	electricity	0.03		0.04		0.04	
	gas	0.04		0.03		0.03	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.08		0.07		0.07	

SC	biomass	0.08	61.4	0.08	45.8	0.09	40.8
	coal	0.02		0.02		0.01	
	electricity	0.34		0.33		0.33	
	gas	0.12		0.12		0.12	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.41		0.42		0.40	
SD	biomass	0.00	5.8	0.00	4.1	0.00	-3.2
	coal	0.00		0.00		0.00	
	electricity	0.05		0.05		0.05	
	gas	0.03		0.03		0.03	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.09		0.09		0.09	
TN	biomass	0.06	75.9	0.06	59.8	0.07	52.9
	coal	0.05		0.05		0.04	
	electricity	0.44		0.44		0.43	
	gas	0.19		0.19		0.19	
	hydrogen	0.01		0.01		0.00	
	refined liquids	0.56		0.57		0.54	
TX	biomass	0.07	601.5	0.07	494.1	0.11	462.8
	coal	0.05		0.05		0.03	
	electricity	1.69		1.71		1.65	
	gas	2.29		2.27		2.27	
	hydrogen	0.02		0.02		0.01	
	refined liquids	5.24		5.21		5.19	
UT	biomass	0.00	58.2	0.00	38.4	0.00	27.2
	coal	0.02		0.02		0.01	
	electricity	0.15		0.16		0.15	
	gas	0.15		0.14		0.14	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.23		0.23		0.22	
VA	biomass	0.08	75.8	0.08	65.3	0.08	58.1
	coal	0.05		0.04		0.03	
	electricity	0.48		0.46		0.45	
	gas	0.21		0.21		0.20	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.63		0.64		0.59	
VT	biomass	0.01	4.5	0.01	3.9	0.01	-0.4
	coal	0.00		0.00		0.00	
	electricity	0.03		0.03		0.03	

	gas	0.01		0.01		0.01	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.07		0.07		0.06	
WA	biomass	0.10	72.7	0.10	66.5	0.10	46.6
	coal	0.00		0.00		0.00	
	electricity	0.44		0.46		0.46	
	gas	0.15		0.14		0.15	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.78		0.76		0.65	
WI	biomass	0.08	78.1	0.08	57.2	0.08	39.2
	coal	0.02		0.02		0.02	
	electricity	0.32		0.28		0.28	
	gas	0.31		0.28		0.28	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.45		0.45		0.41	
WV	biomass	0.01	85.7	0.01	48.4	0.01	26.5
	coal	0.04		0.04		0.03	
	electricity	0.15		0.14		0.14	
	gas	0.13		0.13		0.13	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.16		0.16		0.16	
WY	biomass	0.00	59.7	0.00	26.2	0.00	15.8
	coal	0.03		0.02		0.02	
	electricity	0.07		0.08		0.10	
	gas	0.09		0.09		0.09	
	hydrogen	0.00		0.00		0.00	
	refined liquids	0.11		0.11		0.11	

**A) County-level PM_{2.5}-attributable deaths,
“Existing Climate Policies” relative to “Baseline”, 2030**



**B) County-level PM_{2.5}-attributable deaths,
“All-In Climate Policies” relative “Baseline”, 2030**

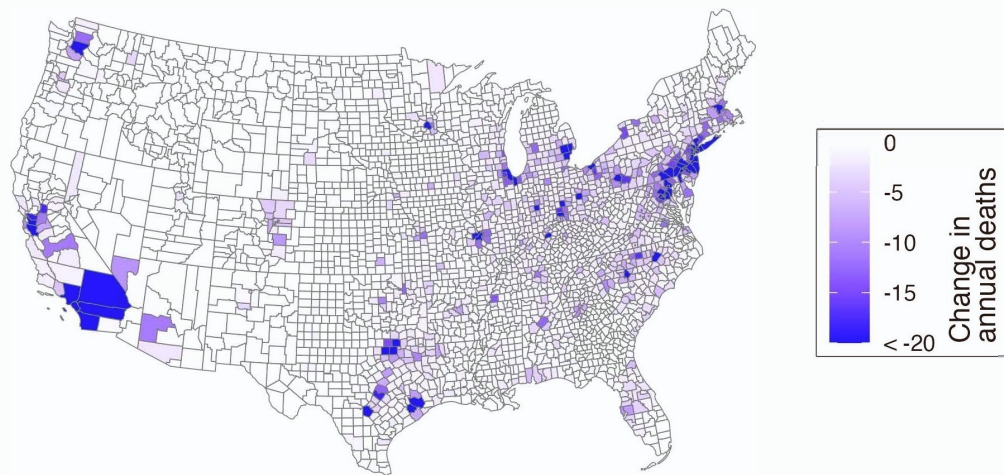


Figure S5. County-level absolute changes of deaths and death rates in the climate policies scenarios relative to the “Baseline” scenario. Panels A) and B) show the county-level absolute reductions in PM_{2.5}-attributable deaths in 2030 in the “Existing Climate Policies” and the “All-In Climate Policies” scenarios, respectively.

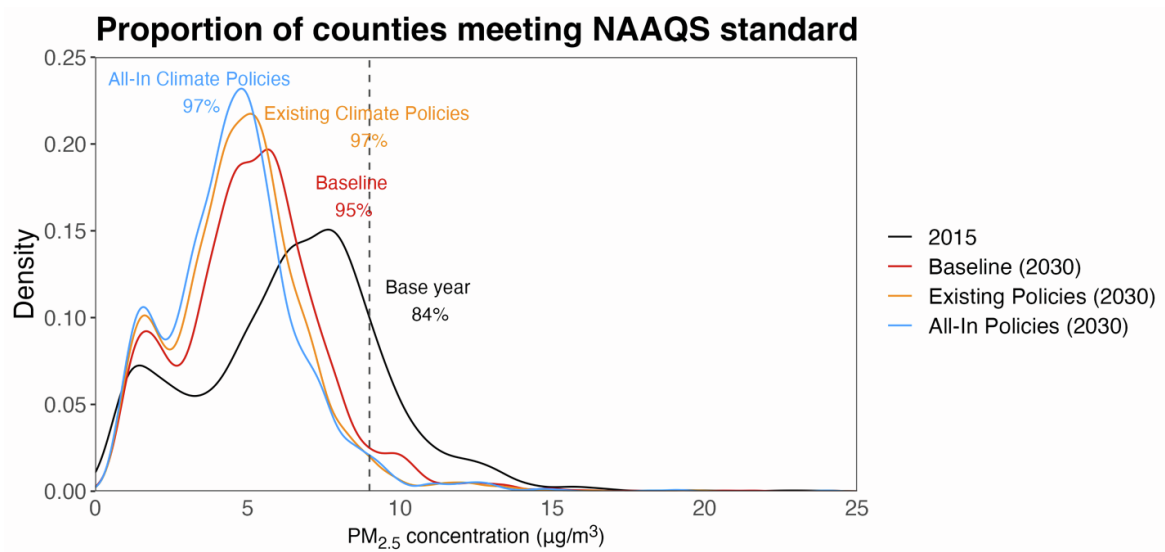


Figure S6. Proportion of 3108 US counties meeting the National Ambient Air Quality Standards (NAAQS) for $PM_{2.5}$ concentration under different policy scenarios for the year 2030, compared to the base year of 2015. The black line represents the 2015 $PM_{2.5}$ concentration distribution, with 84% of counties meeting the NAAQS. Projections for 2030 show that 95% of counties would meet the standard under the “Baseline” scenario (red), 97% under the “Existing Climate Policies” scenario (orange), and 97% under the “All-In Climate Policies” scenario (blue). The vertical dashed line indicates the NAAQS threshold for $12 \mu g/m^3$.

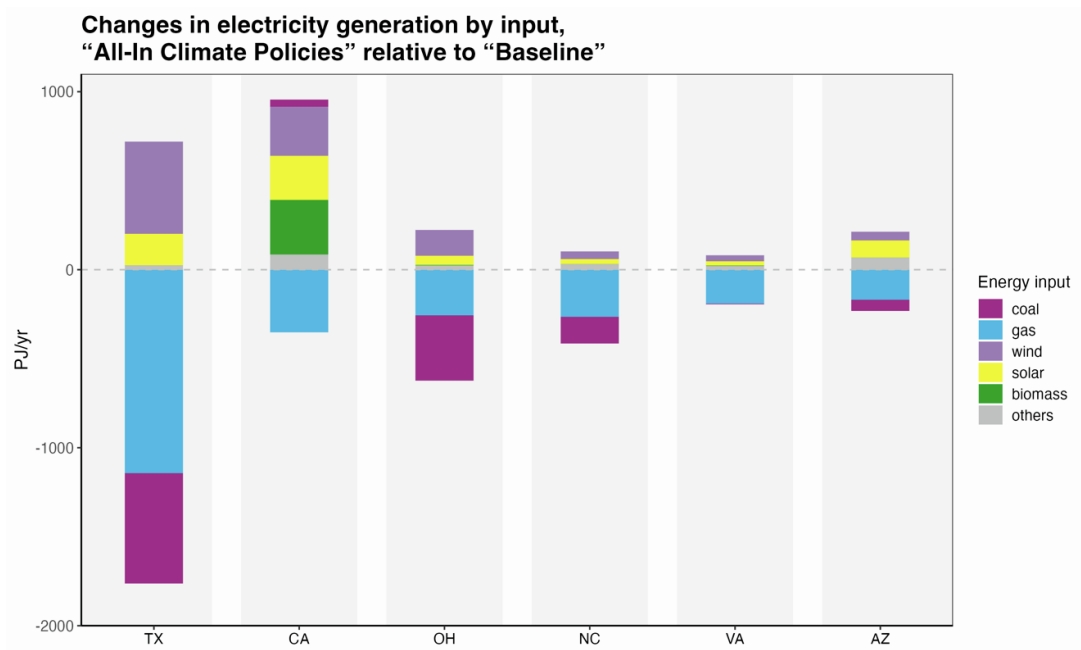


Figure S7. Changes in energy input in electricity generation by state, in the “All-in Climate Policies” scenario relative to the “Baseline scenario” scenario, in 2030. Positive values indicate increase in this energy input, and negative values indicate decrease in this energy input.

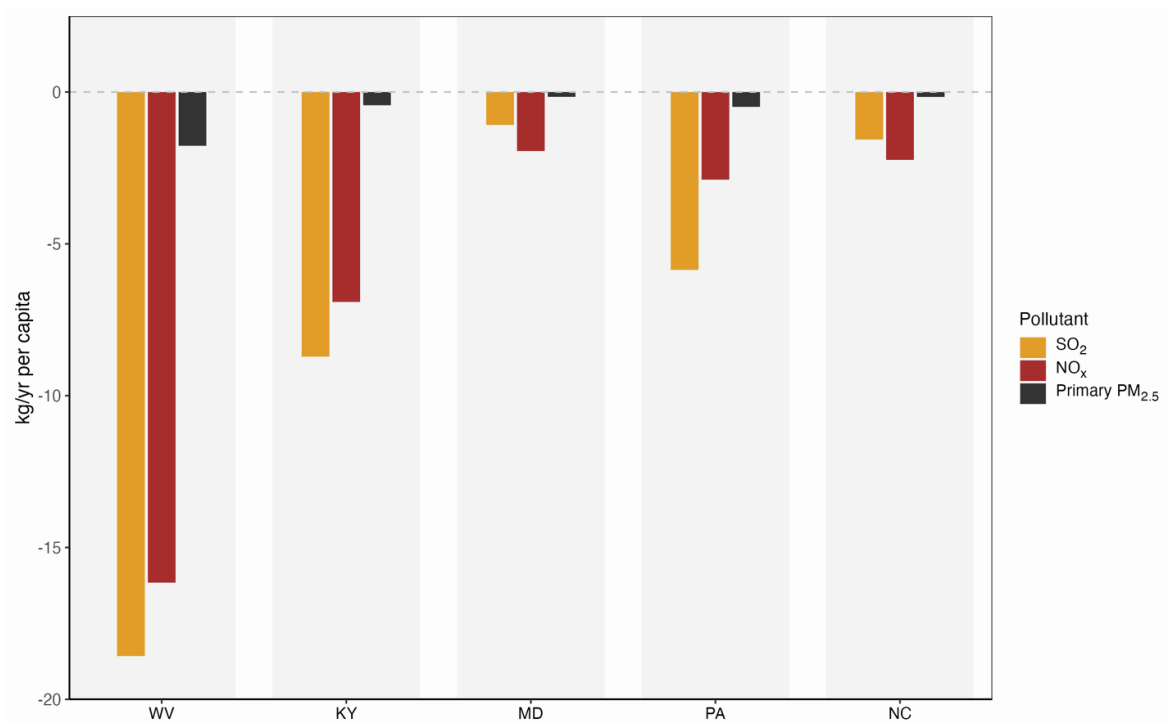


Figure S8. State-level changes in air pollutant emissions in states surrounding Virginia, in the “All-In Climate Policies” scenario relative to the “Baseline scenario”, in 2030.

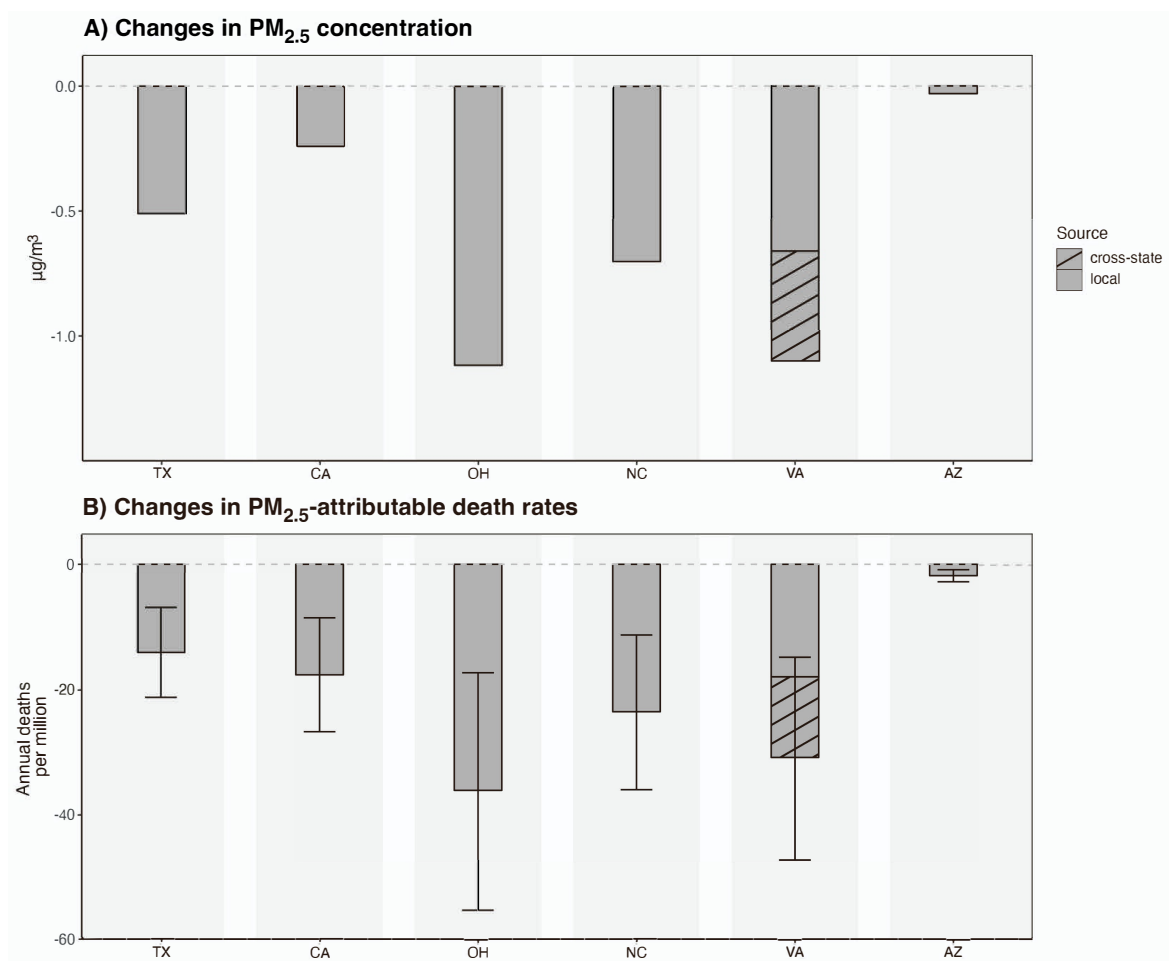


Figure S9. State-level changes in ambient PM_{2.5} concentrations and health impacts in 2030 in the “All-In Climate Policies” scenario relative to the “Baseline” scenario. Panel A) shows the changes in ambient PM_{2.5} concentrations. Panel B) shows the changes in PM_{2.5}-attributable death rate as a measure for health risks; the error bars show the health impact assessment results with upper and lower bounds of concentration-response coefficients. The shaded areas indicate the cross-state transport effects from states surrounding Virginia (PA, NC, MD, KY, and WV) to Virginia.

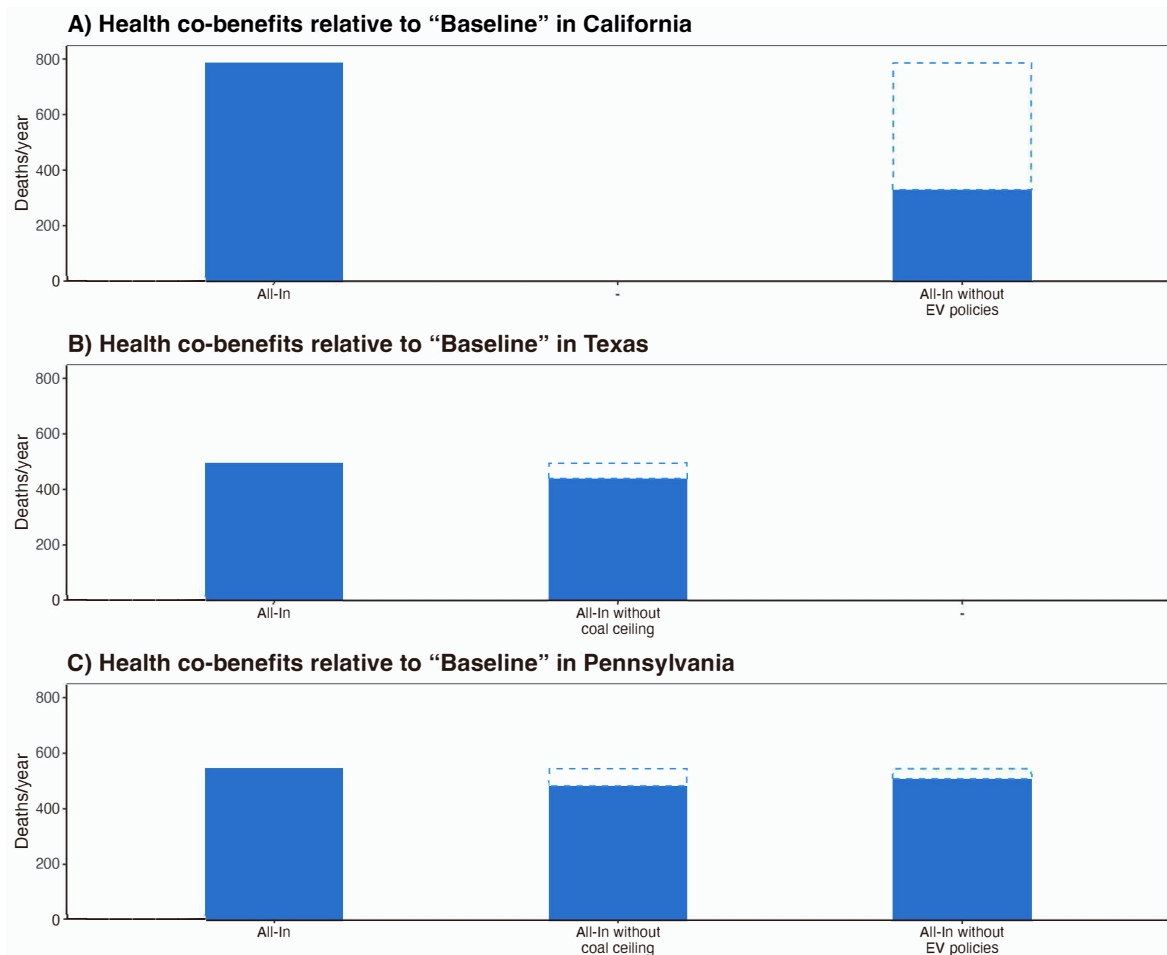


Figure S10. Health co-benefits from state-level isolated actions in CA, TX, and PA in 2030. In each row, health co-benefits in three scenarios are shown. The first bar shows the health co-benefits in the All-In Policies scenario; the second bar shows the health co-benefits when all policies in the All-In Policies scenario are implemented, but the coal ceiling policies are not; the third bar shows the health co-benefits when all policies in the All-In Policies scenario are implemented, but the EV policies are not. Dashed portions of the bars show the reductions of health co-benefits in the scenarios.

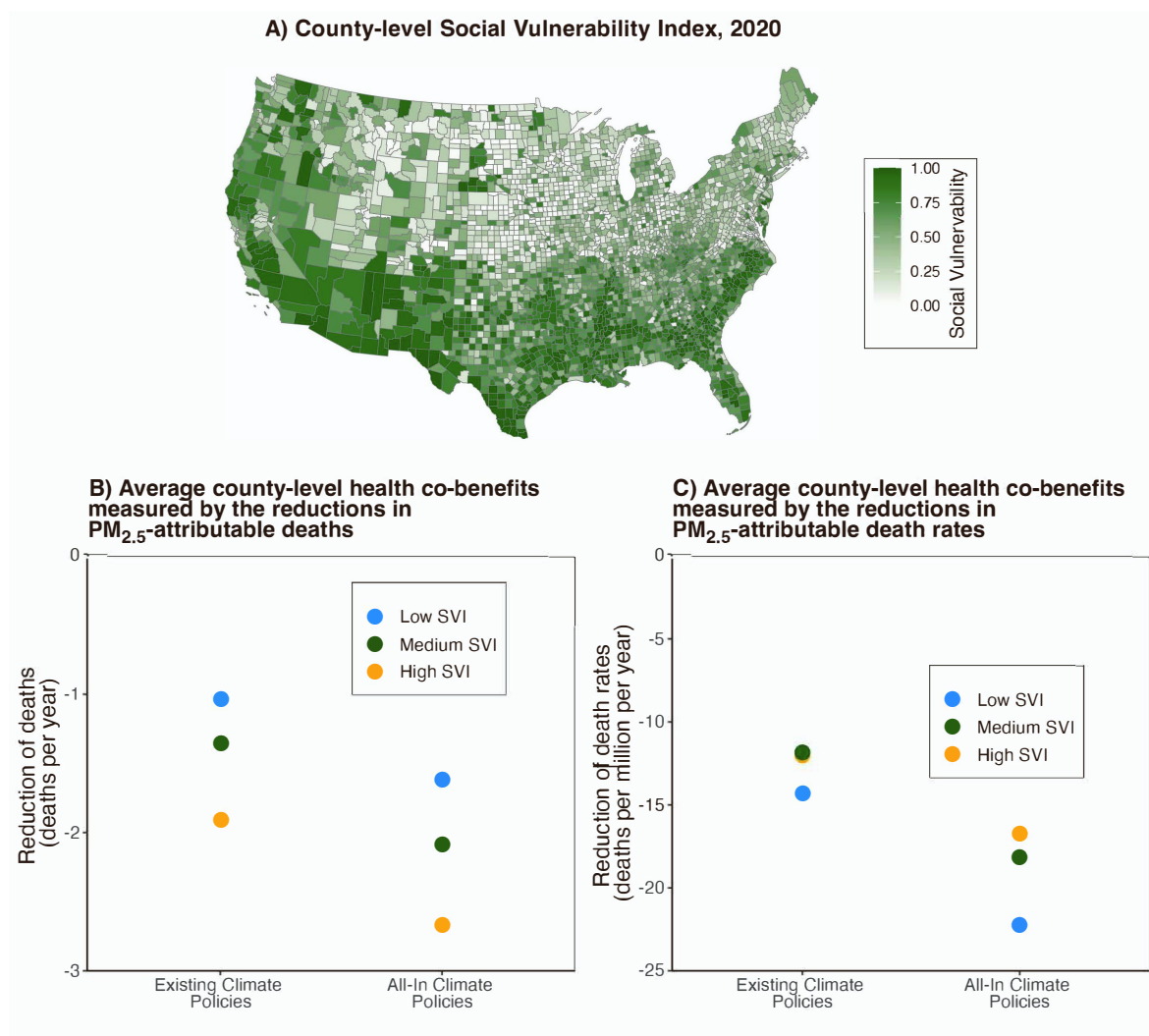
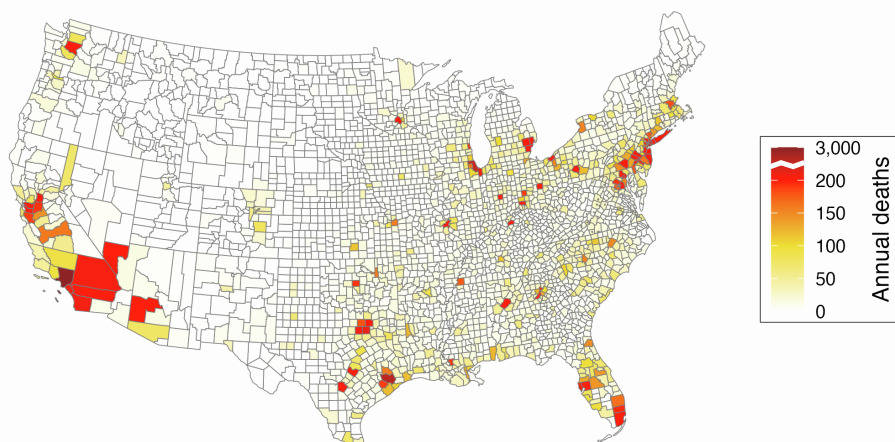


Figure S11. County-level social vulnerability and health co-benefits from climate policies by social vulnerability level. Panel A) shows the county-level Social Vulnerability Indices in 2020. (Source: <https://www.atsdr.cdc.gov/placeandhealth/svi/index.html>) Panel B) shows the average county-level health co-benefits, measured by reductions in $PM_{2.5}$ -attributable deaths, under "Existing Climate Policies" and "All-In Climate Policies" for counties with low (blue), medium (green), and high (yellow) SVI, in 2030. Panel C) shows the average county-level health co-benefits, measured by reductions in $PM_{2.5}$ -attributable death rates (deaths per million people per year), across the same policy scenarios and SVI groups, 2030.

14. County-level PM_{2.5}-attributable deaths in the Baseline scenario using 1 km x 1 km downscaling

a) County-level PM_{2.5}-attributable deaths, NEI, 2015



b) County-level PM_{2.5}-attributable deaths, NEI + NEMO, 2015

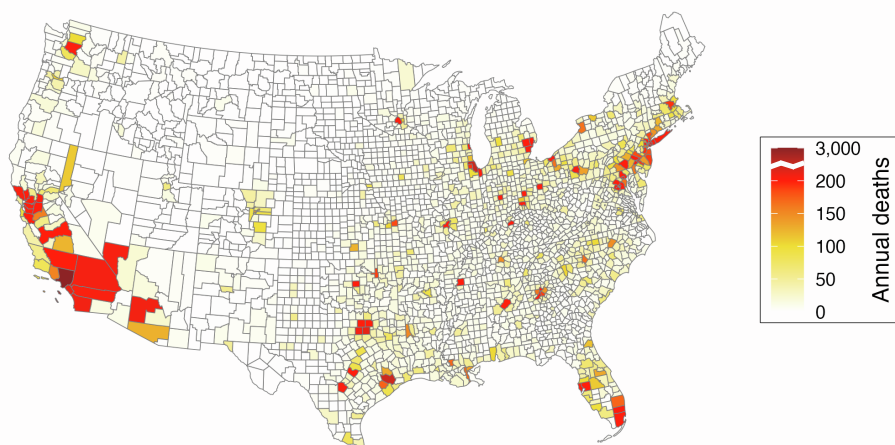


Figure S12. County-level PM_{2.5}-attributable deaths in the Baseline scenario in 2030 using two downscaling approaches. Panel A) shows the evaluation of PM_{2.5}-attributable deaths in 2015 using GCAM air pollutant output for emissions, NEI pattern for downscaling, and InMAP for air quality modeling. Panel B) shows the evaluation of PM_{2.5}-attributable deaths in 2015 using GCAM air pollutant output for emissions, combined NEI and NEMO pattern for downscaling, and InMAP for air quality modeling. In Panel B), we consider an hybrid downscaling approach where we use NEI to downscale power sector emissions to 12 km by 12 km resolution and use Neighborhood Emission Mapping Operation (NEMO) database to downscale the emissions from building, transportation and non-point sources in the industrial sector to 1km by 1km resolution.

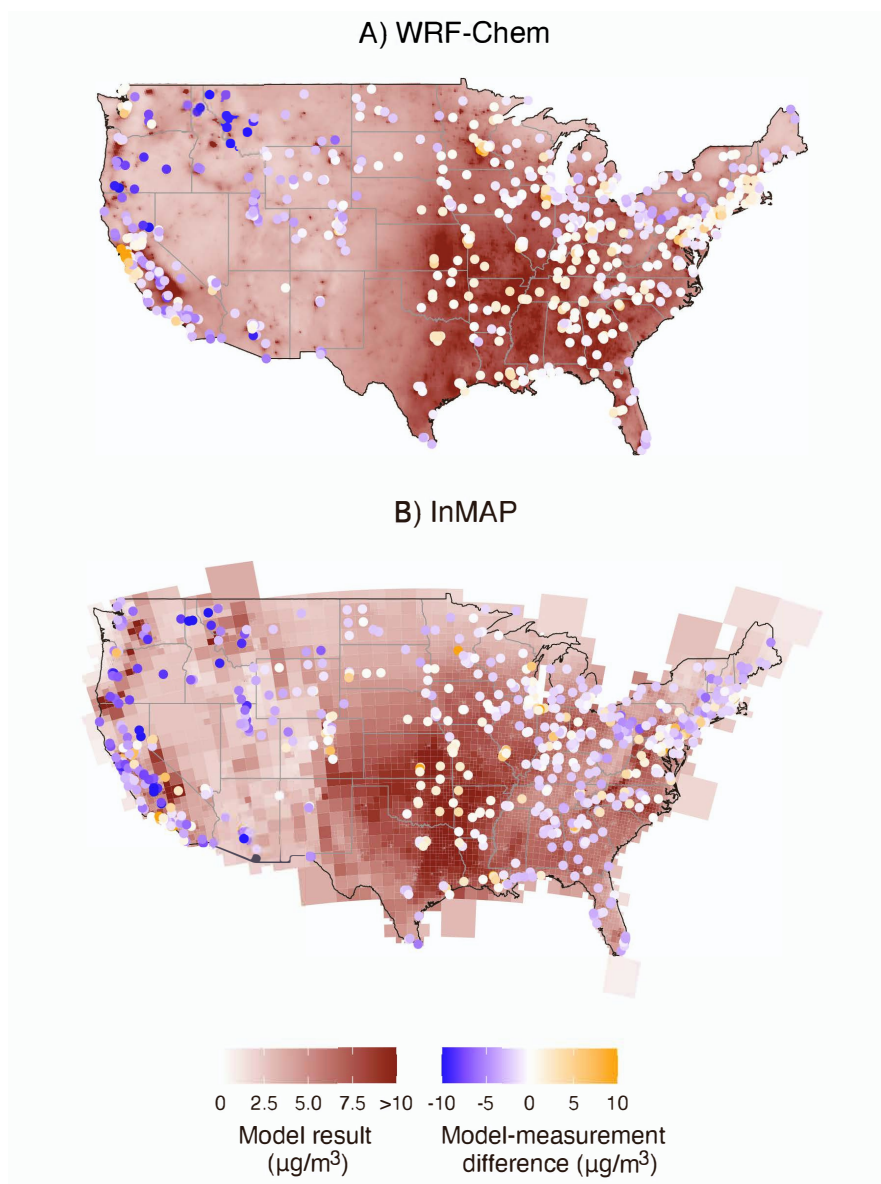


Figure S13. Comparison of model evaluation of $\text{PM}_{2.5}$ concentration using a full-form air quality model (WRF-Chem) and a reduced-complexity model (InMAP). In both simulations, 2017 emissions data from the National Emissions Inventory (NEI) are used as input because it is the latest NEI data that is close to 2015. The simulation results are compared with 2017 $\text{PM}_{2.5}$ concentration observations. Panel A) shows the evaluation of $\text{PM}_{2.5}$ simulation using 2017 NEI emissions data and WRF-Chem. Panel B) shows the evaluation of $\text{PM}_{2.5}$ simulation using 2017 NEI emissions data and InMAP. Both models show an underestimation of $\text{PM}_{2.5}$ concentration in the West Coast.

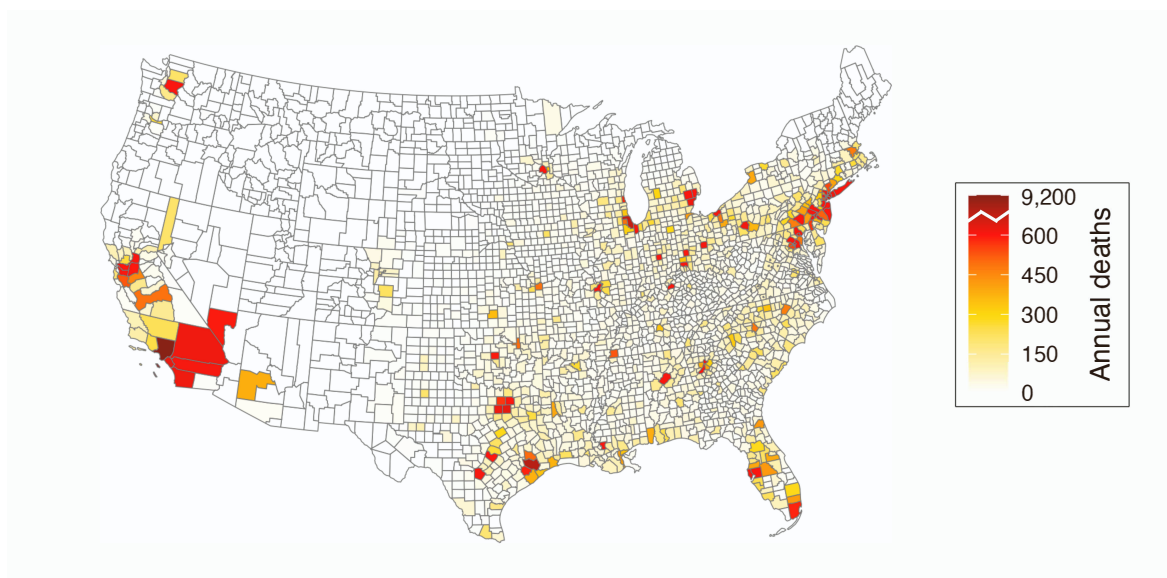


Figure S14. County-level PM_{2.5}-attributable deaths in 2015 using health impact assessment function from the Global Exposure Mortality Model.

Supplemental References

1. Markandya, A., Sampedro, J., Smith, S.J., Van Dingenen, R., Pizarro-Irizar, C., Arto, I., and González-Eguino, M. (2018). Health co-benefits from air pollution and mitigation costs of the Paris Agreement: a modelling study. *Lancet Planet. Health* 2, e126–e133. [https://doi.org/10.1016/S2542-5196\(18\)30029-9](https://doi.org/10.1016/S2542-5196(18)30029-9).
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